Development of an Integrated Management System for the Cagayan Provincial Health Office - Animal Bite Treatment Center

A Thesis Presented to
the Faculty of the Graduate School
Cagayan State University - Carig Campus
Tuguegarao City

In Partial Fulfillment of the
Academic Requirements for the degree
Master of Science in Information Technology

By:

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APPROVAL SHEET

In partial fulfillment of the academic requirements for the degree, Master of Science in Information Technology, this manuscript, capstone project entitled **"Development of an Integrated Management System for the Cagayan Provincial Health Office - Animal Bite Treatment Center"** has been prepared and submitted by Maverick M. Ballesteros hereby recommended for oral defense.

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ACKNOWLEDGEMENT

First and foremost, I would like to express my deepest gratitude to the Almighty God for giving me the strength, wisdom, and perseverance throughout the completion of this academic work.

I would like to extend my sincere appreciation to the Graduate School of Cagayan State University – Carig Campus for providing the academic environment and opportunity that made this study possible. I am especially grateful to my adviser, Dr. Edison D. Bravo, for his valuable guidance and support, and to the proposal-defense panel for their constructive comments and recommendations that helped improve this study.

I would also like to express my appreciation to the public health personnel whose work in animal bite treatment, patient follow-up, and reporting provided the practical foundation for this project. Their dedication inspired the emphasis of this system on complete records, consistent schedules, and accessible information.

My sincere gratitude also goes to my family, friends, colleagues, and everyone who provided encouragement and support throughout the preparation of this system and manuscript. Their understanding and assistance contributed greatly to the completion of this academic requirement.

Lastly, I acknowledge everyone who, in one way or another, contributed to the completion of this study. Above all, I remain grateful for the opportunity to learn and develop through this work and take full responsibility for its implementation, interpretations, and remaining limitations.

The Researcher

DEDICATION

This work is dedicated to my family, teachers, and colleagues whose encouragement and support continue to inspire me in the pursuit of learning and public service. It is also dedicated to the health personnel who tirelessly care for individuals affected by animal bite incidents. May this work and the responsible use of information technology support their daily efforts and contribute to more organized and responsive public health services.

The Researcher

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LIST OF ABBREVIATIONS

Table 1. Abbreviations used in the manuscript

Abbreviation	Meaning
ABTC	Animal Bite Treatment Center
API	Application Programming Interface
DOH	Department of Health
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HIPO	Hierarchical Input Process Output
IEC	International Electrotechnical Commission
IPO	Input Process Output
ISO	International Organization for Standardization
MLGU	Municipal Local Government Unit
PEP	Post-exposure prophylaxis
PHO	Provincial Health Office
PLGU	Provincial Local Government Unit
QR	Quick Response
RHU	Rural Health Unit
SDLC	Software Development Life Cycle
WHO	World Health Organization

ABSTRACT

Title : **Development of an Integrated Management System for the Cagayan Provincial Health Office - Animal Bite Treatment Center**

Name : **Maverick Ballesteros**
Course : **Master of Science in Information Technology**
Adviser : **Dr. Edison D. Bravo**

This study aimed to develop an Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center to support the management of animal bite cases, rabies immunization records, vaccination schedules, and provincial reporting. Animal bite and rabies-related information from participating Animal Bite Treatment Centers (ABTCs) and Rural Health Units (RHUs) must be organized into the reporting categories required by the Department of Health (DOH). The study employed a developmental research approach and Agile principles in designing and developing the system. The application was developed using Vue 3 for the user interface, Laravel 10 for the application programming interface, and a relational database for data management.

The developed system centralizes client and exposure records, case assessment, vaccination scheduling and monitoring, vaccine availability, human rabies records, and reporting functions. It allows existing client information and previous vaccination records to be retrieved using the client's PhilHealth number and supports multiple exposure records for the same client. Vaccination records can also be continued across participating centers while maintaining restrictions on the modification of records created by other centers. The system provides QR-based record access, patient-accessible vaccination history, online pre-registration, and automated email reminders sent one day before a scheduled vaccination. Dashboard functions provide filtered summaries of animal bite cases, exposure categories, vaccination status, sex, age group, biting animals, and human rabies records.

For provincial reporting, the system automatically aggregates encoded records from participating centers and generates the required Rabies Report Form according to the prescribed DOH reporting structure. Center-level accomplishment reports are also generated from recorded service data. An ISO/IEC 25010:2011-based instrument was prepared to evaluate the software quality of the developed system.

The developed system provides a centralized approach to animal bite case management, vaccination monitoring, cross-center treatment continuity, and provincial rabies reporting. Further implementation should include continued validation of generated reports, reminder delivery, access restrictions, data accuracy, and system maintenance.

Keywords: *integrated management system, animal bite treatment center, rabies reporting, vaccination monitoring, post-exposure prophylaxis, public health information system, Agile development*

CHAPTER I

THE PROBLEM AND ITS BACKGROUND

1.1 Introduction

Animal bite treatment involves information that develops across several stages of care. An initial encounter records the identity of the patient, circumstances of the exposure, characteristics of the biting animal, and the recommended course of treatment. Subsequent visits add vaccination records, treatment updates, and follow-up information. Maintaining continuity among these records is important because treatment may extend across several scheduled visits and, in some cases, may be continued at a different treatment center.

Rabies remains a preventable public health concern for which timely assessment, appropriate post-exposure prophylaxis, and continued monitoring are essential. The World Health Organization (WHO, 2026) emphasizes prevention, access to appropriate care, and surveillance as important components of rabies control. Information management supports these activities by helping health personnel maintain accurate records, monitor vaccination schedules, retrieve previous treatment information, and organize data required for public health reporting. This study focuses on these information-management functions rather than on evaluating the clinical effectiveness of rabies treatment.

In the health sector, information systems play a critical role in managing health records, monitoring public health cases, and ensuring timely reporting. Health Information Systems (HIS) support the systematic collection, processing, storage, and dissemination of health-related data necessary for planning, monitoring, and policy formulation (World Health Organization [WHO], 2022). The utilization of IT in public health settings has been shown to improve data accuracy, reduce manual workload, and enhance the accessibility and continuity of health records.

Rabies remains a serious public health concern, particularly in developing countries, where animal bite incidents continue to pose significant risks to human health. Effective monitoring of animal bite cases and timely administration of post-exposure prophylaxis are essential to prevent rabies-related mortality. However, several studies have reported that many local health facilities still rely on manual and paper-based methods for recording animal bite cases and tracking vaccination schedules. These practices often result in delayed reporting, data inconsistencies, difficulty in retrieving patient records, and challenges in monitoring vaccination compliance (Sarkar et al., 2020).

In the Philippine setting, Provincial and Municipal Health Units are responsible for documenting animal bite cases, administering rabies post-exposure prophylaxis, and submitting consolidated reports to higher health authorities. Despite ongoing initiatives promoting digital governance, many health facilities continue to experience challenges due to manual record-keeping and fragmented data management systems. These challenges include time-consuming client registration processes, increased workload for health personnel, limited visibility of real-time provincial data, missed vaccination schedules, and delays in report consolidation and submission (Department of Health [DOH], 2021).

Recent studies emphasize the importance of integrating digital technologies into public health information systems to address these operational challenges. Centralized information systems enable efficient data sharing across different administrative levels, while automation minimizes human error and improves the timeliness and reliability of reports (Garcia et al., 2022). Furthermore, the application of Quick Response (QR) code technology in healthcare settings has been shown to improve patient identification and facilitate faster retrieval of health records, thereby reducing transaction time and improving service efficiency without replacing clinical judgment (Rahman & Islam, 2023).

In the Philippines, Republic Act No. 9482, or the Anti-Rabies Act of 2007, provides a national framework for rabies prevention and control involving national government agencies, local government units, and health institutions. Animal Bite

Treatment Centers and Rural Health Units form part of the service network through which animal bite cases are assessed, treated, monitored, and reported. Previous studies involving the Philippine Animal Bite Treatment Center network have also identified operational and patient-access concerns that demonstrate the importance of organized service delivery and continuity of care (Amparo, Jayme, Roces, Quizon, Mercado, et al., 2018; Amparo, Jayme, Roces, Quizon, Villalon, et al., 2018). Within the Province of Cagayan, animal bite and rabies-related information originates from participating treatment centers and must eventually be organized for provincial monitoring and reporting. Information on animal bite cases, exposure classifications, post-exposure prophylaxis, biting animals, and human rabies cases is required for the preparation of reports following the format prescribed by the Department of Health. When treatment information and reporting data are maintained through separate records or manually prepared summaries, retrieval, monitoring, and consolidation require additional effort and may involve repeated encoding and checking of the same information.

Continuity of treatment presents another information-management concern. A patient may receive the initial assessment and vaccination at one center but continue succeeding doses at another participating center. In such situations, the receiving health personnel need access to the relevant treatment history while the integrity of records created by the originating center must still be maintained. The use of a shared vaccination record, supported by QR-based record retrieval and a printable vaccination slip, can provide patients and receiving centers with accessible treatment information while allowing the vaccination history to remain connected across participating facilities. The same patient may also experience another animal exposure at a later time, requiring a new case record while retaining access to previous vaccination information.

These conditions establish the need for a centralized information environment in which patient records, exposure information, vaccination schedules, treatment history, and reporting data can remain connected. A shared system can reduce repeated encoding, make vaccination histories easier to

retrieve, support monitoring of scheduled doses, preserve continuity when treatment is continued across centers, and provide structured data for provincial-level reporting.

The study therefore focuses on the development of an Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center. The system is intended to support the organized management of animal bite treatment information across participating centers while providing the Provincial Health Office with consolidated information for monitoring and reporting. Its development is also evaluated using defined software quality criteria to determine the suitability of the application for its intended operational setting.

1.2 Conceptual Framework

This section discusses the main concept and research paradigm of the study.

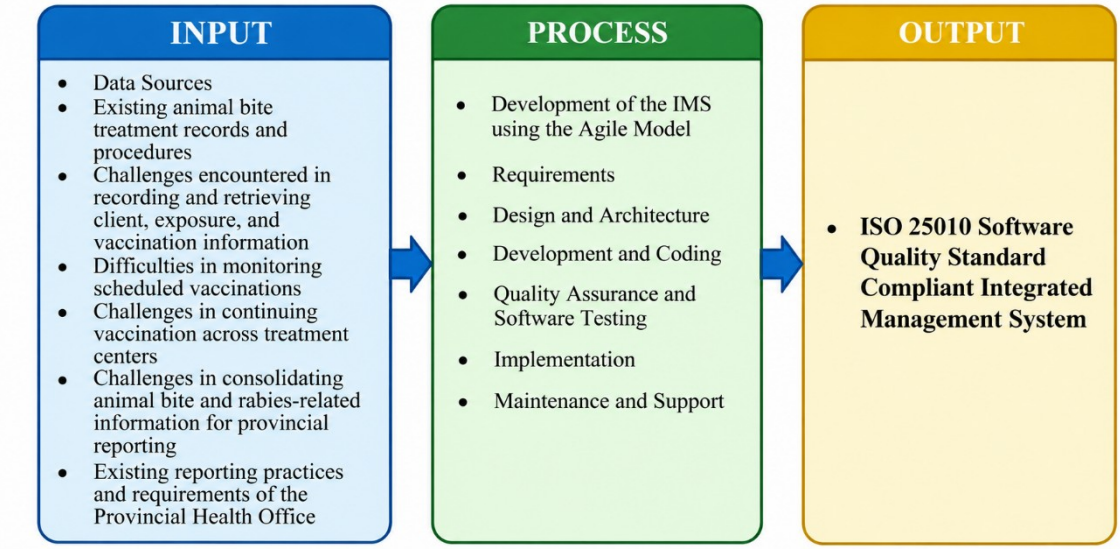


Figure 1.1. Conceptual Framework

As shown in Figure 1.1, the input of the study consists of data sources related to existing animal bite treatment records and procedures, challenges in recording and retrieving client, exposure, and vaccination information, difficulties in

monitoring scheduled vaccinations, concerns in maintaining continuity of vaccination across treatment centers, and challenges in consolidating animal bite and rabies-related information for provincial reporting. Existing reporting practices and requirements of the Provincial Health Office also serve as important inputs in identifying the functional and operational needs of the proposed system.

These inputs are used in the development of the Integrated Management System through the Agile Model. The development process includes requirements gathering, system design and architecture, development and coding, quality assurance and software testing, implementation, and maintenance and support. The Agile approach allows the development activities to be carried out iteratively, enabling revisions and improvements to be incorporated as requirements and operational concerns are identified during the development process.

The output of the framework is an ISO 25010 Software Quality Standard Compliant Integrated Management System. The developed system is evaluated based on the ISO 25010 software quality characteristics, including functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

1.3 Research Paradigm

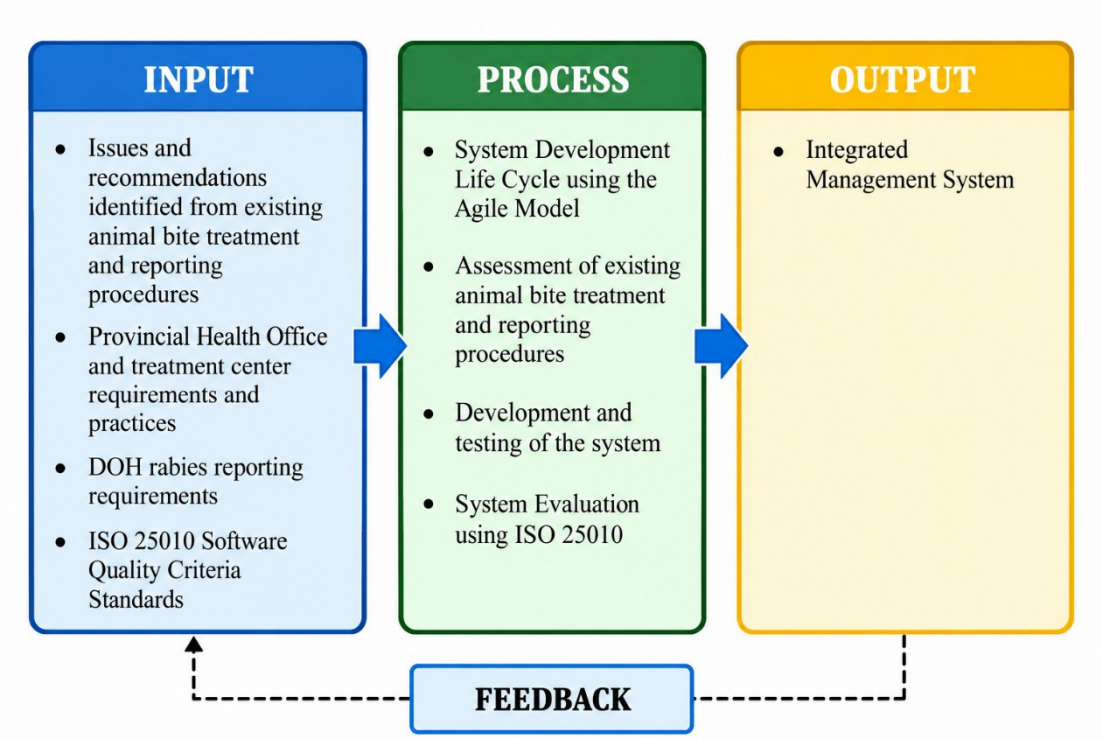


Figure 1.2. Research paradigm

As shown in Figure 1.2, the inputs of the study include the issues and recommendations identified from the existing animal bite treatment and reporting procedures, the requirements and practices of the Provincial Health Office and participating treatment centers, the Department of Health rabies reporting requirements, and the ISO 25010 software quality criteria standards. These inputs serve as the basis for determining the functional, operational, and quality requirements of the Integrated Management System.

The development process follows the Agile Model, which supports iterative planning, development, testing, and improvement of the system based on identified requirements and feedback. The process also includes the assessment of existing animal bite treatment and reporting procedures, development and testing of the system, and system evaluation using ISO 25010.

Through these inputs and processes, an Integrated Management System was developed to support animal bite case management, vaccination

monitoring, cross-center continuity of treatment, and provincial reporting. The feedback mechanism allows identified concerns, user observations, and evaluation results to be considered for further refinement and improvement of the system.

1.4 Objectives of the Study

1.4.1 General Objective

To develop an Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center that supports animal bite case management, vaccination monitoring and continuity of treatment, and provincial reporting.

1.4.2 Specific Objectives

Specifically, the study seeks to achieve the following objectives:

1. To identify the challenges encountered by health personnel in using the existing manual process of animal bite case recording and rabies immunization management.
2. To design and develop an Integrated Animal Bite Treatment Center Information System with QR Based Vaccination Schedule Retrieval and Automated E-mail Notifications for Provincial and Municipal Health Units.
3. To determine the compliance of the developed system with the ISO/IEC 25010 Software Quality Standards in terms of:

    a. Functional Suitability

    b. Performance Efficiency

    c. Compatibility

    d. Usability

    e. Reliability

f. Security

g. Maintainability

h. Portability

1.5 Scope and Limitation of the Study

The study covers the development and evaluation of an Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center. The system manages client and contact information, animal bite and exposure details, vaccination schedules and administration records, treatment-center information, vaccine availability, human rabies records, user accounts, report signatories, and selected application settings. Public-facing services include client pre-registration, viewing of treatment-center information, and verification of vaccination history. Staff functions include case assessment, record retrieval, vaccination monitoring, and dose documentation, while administrator functions include user and center management and access to provincial-level reports.

The system is intended for use within the Provincial Government of Cagayan and participating Animal Bite Treatment Centers, Rural Health Units, and related health facilities. Intended users include personnel involved in patient intake, vaccination documentation, monitoring, report preparation, and system administration.

The study focuses on information management, vaccination monitoring, continuity of treatment, and reporting functions. QR codes are used to assist in retrieving exposure-related records but do not replace staff verification of patient identity. Treatment-related classifications and system rules are intended to support record management and are not presented as substitutes for professional medical judgment.

Automated email reminders depend on the availability of a valid patient email address, the configured server scheduler, and the external email service

used by the system. The study does not cover a complete vaccine inventory or stock-ledger system, PhilHealth claims processing, laboratory information management, or offline operation. Vaccine availability indicates whether a center has vaccine available but does not represent audited stock quantities.

The technical description of the system reflects the implementation available during the conduct and documentation of the study. Evaluation findings are limited to the data and testing activities conducted. The study does not claim results from assessments, security tests, performance tests, interviews, or pilot evaluations that were not completed or documented.

1.6 Significance of the Study

The study will benefit the following:

Clients. Public pre-registration and information services can support preparation for a visit and access to recorded vaccination information. These functions should accommodate individuals who lack reliable internet access or email; assisted on-site registration remains important.

Health Personnel. The application groups intake, assessment, history, and vaccination-related tasks around a common record structure. This can reduce the need to search separate record sources, although the amount of time saved must be established empirically.

Provincial Health Office. The project provides an organized basis for reviewing center-linked cases and preparing summary reports. Its value lies in making operational information available through connected workflows. Decisions based on those summaries still depend on complete source records and agreed reporting definitions.

Researchers and Developers. The project provides an applied example of translating local health-service requirements into a web application with documented limitations and an explicit evaluation plan.

Treatment-Center Administrators. Maintenance screens provide a consistent place to manage center information, vaccine availability, user accounts, and other operational records needed for treatment-center administration.

1.7 Definition of Terms

Administered Dose. A vaccination dose documented as given, including the applicable date of administration and service-center reference.

Agile Development. An iterative development approach that uses prioritized requirements, incremental development, review, and adaptation throughout the system development process.

Animal Bite Treatment Center (ABTC). A health facility responsible for the assessment and management of animal bite clients within the context of the study.

Assessment. The recording and review of demographic, incident, exposure, and treatment-related information by authorized personnel.

Audit Record. A stored record of an application transaction or change that identifies the affected record and the responsible user, where implemented.

Client. A person whose identity and contact information are maintained in the system. A client may have more than one recorded animal bite or exposure incident.

Exposure Episode. A specific animal bite or exposure record linked to assessment information and, when applicable, a vaccination schedule.

Integrated Management System. The developed application that connects related records and functions for registration, case assessment, vaccination monitoring, reporting, and administration.

ISO/IEC 25010. A software product quality model used in this study for evaluating the developed system. The study adopts the 2011 edition, which includes the eight quality characteristics used in the proposal instrument.

Post-Exposure Prophylaxis (PEP). Preventive care provided after a potentially rabies-related exposure, as determined by qualified health personnel and applicable clinical guidance.

Pre-registration. The advance submission of client demographic information before staff verification and formal assessment at a treatment center.

QR Code. A machine-readable code used by the system to assist in retrieving an exposure-related record.

Scheduled Dose. A planned vaccination date recorded in the system. It is distinct from the actual date on which the vaccine is administered.

User View. The interface and workflow available to a particular user role, including the inputs, actions, and outputs relevant to that role.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a review of related literature and studies gathered from recent scholarly articles, technical reports, and prior research related to health information systems, animal bite case management, rabies post-exposure prophylaxis (PEP) scheduling, QR code technology, automated notification systems, and software quality evaluation. Foreign and local studies published within the last five years and selected foundational standards relevant to the study are discussed to provide theoretical and empirical support for the development of the proposed Integrated Animal Bite Treatment Center Information System with QR-Based Vaccination Schedule Retrieval and Automated E-Mail Notifications.

2.1 Foreign Literature

Agile Development

The Agile Manifesto emphasizes working software, collaboration, and responsiveness to change as important principles in software development (Beck et al., 2001). These principles support a development process in which requirements may be reviewed, prioritized, implemented, and refined through incremental development.

Agile development is relevant to the present study because an integrated health information system involves several interconnected workflows, including client registration, case assessment, vaccination monitoring, reporting, and system administration. An iterative approach allows these functions to be developed and reviewed in manageable increments while incorporating changes that may arise during development.

Schwaber and Sutherland (2020) describe Scrum as an empirical framework based on transparency, inspection, and adaptation. These concepts further support regular review and improvement during software development. In the present study, Agile principles provide the basis for organizing development activities and adapting system functions according to identified operational requirements. Pasted markdown

Digital Health and Information Management

Digital health technologies are increasingly used to support health-service delivery, information management, communication, and monitoring. The World Health Organization (WHO, 2019) emphasizes that digital health interventions should be considered not only in terms of technical functionality but also in relation to benefits, harms, feasibility, acceptability, resource requirements, and equity.

The WHO global strategy on digital health further emphasizes the importance of coordinating organizational, financial, human, and technological resources in implementing digital health initiatives (WHO, 2021). Digital systems are most useful when responsibilities for data entry, correction, access, system continuity, and maintenance are clearly defined.

These principles are relevant to the present study because the proposed system manages connected information involving client records, animal bite exposure details, vaccination schedules, treatment history, treatment-center information, and reporting data. A centralized information environment can support continuity and accessibility of these records when accompanied by appropriate organizational processes and user responsibilities. Pasted markdown

Evaluation of Digital Health Interventions

WHO (2016) provides guidance on monitoring and evaluating digital health interventions and emphasizes the distinction between technical functionality and evidence of outcomes. A functioning feature may demonstrate that a technical process works, but this is different from establishing an improvement in clinical outcomes, patient attendance, or service efficiency.

This distinction is relevant to the present study because the evaluation of the developed system focuses primarily on software quality and operational functionality. The presence of automated reminders, public information services, or record-retrieval features does not by itself establish improvements in treatment

completion or other health outcomes. Such outcomes require separate measures, appropriate baselines, and follow-up data. Pasted markdown

QR Code Technology and Record Retrieval

DENSO WAVE (n.d.) describes the QR code as a two-dimensional code capable of storing information that can be read using a compatible scanning device. QR codes can provide a convenient means of accessing or locating a record without requiring the manual entry of lengthy identifiers.

In the context of the present study, QR-based retrieval can support access to exposure-related and vaccination information when a patient presents at a participating treatment center. This is particularly relevant when vaccination is continued at a center different from the one where the initial assessment was conducted.

However, QR codes should be regarded as a record-retrieval mechanism rather than as proof of identity. Authorized personnel still need to verify that the retrieved record corresponds to the correct patient before treatment information is updated. Pasted markdown

Rabies Prevention, Surveillance, and Continuity of Care

The WHO Expert Consultation on Rabies provides technical and programmatic guidance for rabies prevention, control, and elimination (WHO, 2018). It highlights the importance of organized information in supporting rabies-related programs and monitoring activities.

The Zero by 30 strategy places the prevention of human rabies within a coordinated international effort involving surveillance, vaccination, organized service delivery, and collaboration among institutions (WHO et al., 2018). Reliable information concerning animal exposure, vaccination, and treatment is necessary to support these activities.

WHO (2024) also identifies animal bites as an important injury and public health concern. This supports the need to document the circumstances of animal exposure, characteristics of the biting animal, and treatment-related information rather than limiting records to vaccination counts alone.

These principles are relevant to the present study because animal bite treatment may involve several encounters over a period of time. Client information, exposure details, vaccination schedules, administered doses, and follow-up records need to remain connected to support continuity of treatment, particularly when succeeding doses are administered at different treatment centers. Pasted markdown

Software Quality

ISO/IEC 25010:2011 provides a software product quality model containing eight quality characteristics: functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability (International Organization for Standardization & International Electrotechnical Commission [ISO/IEC], 2011).

Functional suitability concerns the degree to which software provides functions that meet stated and implied needs. Performance efficiency refers to system performance in relation to the resources used. Compatibility addresses the ability of the software to operate and exchange information within its environment. Usability relates to the degree to which users can use the system effectively and satisfactorily. Reliability concerns consistent system performance under specified conditions. Security refers to the protection of information and access. Maintainability concerns the ease of modifying or improving the software, while portability refers to its ability to operate in different environments.

The 2023 revision of ISO/IEC 25010 introduces changes to the software product quality model. However, the present study adopts the 2011 edition because the evaluation instrument identified in the proposal is based on its eight quality characteristics. Pasted markdown

Technical Foundations of Web-Based Information Systems

Vue supports the development of component-based user interfaces in which visual elements and application functions can be organized into reusable components (Vue.js, n.d.). This structure is useful in web applications that contain multiple user interfaces and workflows.

Laravel provides facilities for application programming interfaces, authentication, and scheduled application tasks. Laravel Sanctum supports authentication mechanisms for API-oriented applications, while Laravel scheduling facilities support recurring automated processes (Laravel, n.d.-a; Laravel, n.d.-b).

MySQL InnoDB provides transactional mechanisms that help maintain consistency when related records must be created or updated together (Oracle, n.d.). Such mechanisms are relevant in systems where client, exposure, schedule, and vaccination records are stored across related tables.

Programmatic email services also provide mechanisms for sending automated communication from an information system. Microsoft's sendMail documentation illustrates an interface for sending email through an application (Microsoft, n.d.), making email integration relevant to systems that provide patient notifications and vaccination reminders. Pasted markdown

Vaccination Reminders and Follow-up

Jacobson Vann et al. (2018) reviewed patient reminder and recall interventions for immunization and found evidence that such interventions can improve immunization uptake across different populations and communication methods.

The findings support the use of reminder mechanisms as an additional means of encouraging patients to return for scheduled vaccination. In an information system, scheduled vaccination dates should be distinguished from

actual administration dates to allow health personnel to determine whether a scheduled dose has been given, remains pending, or has become overdue.

For the present study, automated email reminders are intended to support vaccination follow-up. However, the effectiveness of these reminders in improving rabies vaccination completion within the Cagayan setting requires separate evaluation using appropriate outcome measures. Pasted markdown

Web Accessibility, Security, and Privacy

The Web Content Accessibility Guidelines provide testable criteria for improving accessibility in web applications, including keyboard access, labeling, visual contrast, and error identification (World Wide Web Consortium [W3C], 2024). These guidelines provide useful considerations in designing web interfaces that may be accessed by users with different needs and abilities.

Security is also an important consideration for applications that process identifying and health-related information. The OWASP Application Security Verification Standard provides structured guidance for evaluating security controls in web applications (OWASP Foundation, 2025). The NIST Privacy Framework provides guidance for identifying and managing privacy-related risks associated with information systems (National Institute of Standards and Technology [NIST], n.d.).

These frameworks provide useful references for considering access controls, privacy, information disclosure, and security practices in the developed system. Their use as reference materials does not, however, automatically establish complete certification or compliance.

2.2 Local Literature

Health-Provider Experience in Animal Bite Treatment Centers

Amparo, Jayme, Roces, Quizon, Villalon, et al. (2018) evaluated operating Animal Bite Treatment Centers from the perspective of health-service providers. Their study documented differences in service costs and operational challenges among the facilities examined.

The findings demonstrate that Animal Bite Treatment Centers operate under different local conditions and may experience different administrative and service-delivery concerns. This supports the need to retain treatment-center information in records and reports so that health personnel and administrators can identify the facility responsible for particular services.

For the present study, center-linked information is particularly relevant because patient assessment, vaccination administration, monitoring, and reporting may involve more than one participating treatment center. Maintaining the identity of the center responsible for a service activity supports the interpretation of records and reports according to the facility that provided the treatment. Pasted markdown

Patient Experience in Animal Bite Treatment Centers

Amparo, Jayme, Roces, Quizon, Mercado, et al. (2018) examined the experiences of animal bite patients through community and Animal Bite Treatment Center surveys conducted in three Philippine provinces. Their study considered treatment-seeking behavior and out-of-pocket expenses among patients.

The findings show that access to animal bite treatment involves concerns beyond the availability of clinical services. Patients may experience financial, transportation, and other practical barriers when obtaining treatment.

The study is relevant to the present research because public information services, vaccination reminders, and accessible treatment records may support communication and preparation for treatment visits. However, such features do not eliminate barriers such as transportation costs, lost working time, or distance from treatment facilities. The system should therefore be viewed as a support mechanism for information management and service organization rather than as a complete solution to all barriers affecting animal bite treatment. Pasted markdown

PhilHealth and Treatment Documentation

PhilHealth introduced an animal bite treatment package that provides historical context for the documentation of animal bite treatment services. PhilHealth Circular No. 015 describes the animal bite treatment package, while related issuances provide supplementary documentation requirements (Philippine Health Insurance Corporation [PhilHealth], 2012a, 2012b, 2012c).

These issuances demonstrate the importance of maintaining complete client and treatment-related information within animal bite treatment services. The present system stores PhilHealth-related identifiers and treatment information as part of its client and case records.

However, the developed system is not intended as an automated PhilHealth claims-submission or benefit-eligibility verification platform. The PhilHealth materials are therefore used as background references supporting the importance of accurate and organized treatment documentation. Pasted markdown

Philippine Rabies-Control Framework

Republic Act No. 9482, also known as the Anti-Rabies Act of 2007, provides the national legislative framework for rabies prevention and control in the Philippines. The law establishes responsibilities involving government institutions,

local government units, and other concerned stakeholders in rabies prevention and control.

This framework provides an important context for the present study because Animal Bite Treatment Centers and other local health facilities participate in the assessment, treatment, monitoring, and documentation of animal bite cases. An organized information system can support these activities by providing structured means for recording, retrieving, monitoring, and reporting treatment-related information.

A regional program briefer from the Department of Health – Cordillera Administrative Region (2019) also illustrates the role of program monitoring in a Philippine rabies-control setting. Although information from another region is not treated as Cagayan-specific evidence, it provides an example of organized monitoring and reporting within a local health program. Pasted markdown

Protection of Personal and Health Information

Republic Act No. 10173, or the Data Privacy Act of 2012, provides the Philippine legal framework for the responsible processing and protection of personal information. Its implementing rules and regulations further provide guidance concerning the collection, processing, storage, access, retention, and disclosure of personal information (Republic Act No. 10173, 2012; National Privacy Commission, 2016).

These requirements are relevant to the present study because the developed system manages identifying and health-related information, including client identity, contact details, animal bite exposure information, vaccination records, and treatment history. Authentication, access controls, appropriate disclosure, and proper handling of patient information are therefore important considerations in system implementation.

Patient-facing functions also require appropriate privacy safeguards. Information displayed through public history-verification functions should be

limited to what is necessary for the intended purpose and should include reasonable means of verifying identifying information before treatment-related records are displayed

2.3 Synthesis

The reviewed foreign and local literature demonstrates the importance of organized information management in supporting animal bite treatment, vaccination monitoring, continuity of care, surveillance, and public health reporting. International literature on digital health emphasizes that successful implementation depends not only on the availability of software but also on organizational processes, user responsibilities, infrastructure, and appropriate evaluation.

The reviewed literature on rabies prevention and control highlights the importance of reliable information concerning animal exposures, vaccination, treatment follow-up, and surveillance. Reminder and recall mechanisms provide support for vaccination follow-up, while QR-code technology offers a practical means of retrieving related records. These mechanisms are particularly relevant when treatment extends across several visits or when patients receive succeeding doses at different treatment centers.

Agile development provides an appropriate approach for developing a system whose requirements involve multiple interconnected workflows and may require refinement throughout implementation. ISO/IEC 25010:2011 provides a structured basis for evaluating the developed system according to functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

Within the Philippine setting, the Anti-Rabies Act of 2007 establishes the broader national context for rabies prevention and control, while the Data Privacy Act of 2012 provides guidance relevant to the protection of personal and health-related information. PhilHealth issuances concerning animal bite treatment further demonstrate the importance of complete treatment documentation. Local

studies involving Animal Bite Treatment Centers also show that both health personnel and patients experience operational, administrative, access, and service-related concerns that should be considered when designing systems intended to support animal bite treatment.

The reviewed literature establishes the importance of record organization, vaccination monitoring, patient follow-up, center-linked information, privacy, software quality, and public health reporting. However, the reviewed materials do not specifically describe an integrated system designed for the Cagayan Provincial Health Office that combines client and exposure management, vaccination scheduling and monitoring, cross-center continuation of treatment, patient-accessible vaccination information, treatment-center and vaccine availability information, automated email reminders, center-level reporting, and consolidated provincial rabies reporting within a single platform.

This identified gap provides the basis for the development of the Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center. The study integrates these functions within a centralized web-based application and evaluates the developed system using the ISO/IEC 25010:2011 software quality model.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the methods and procedures used in the development and evaluation of the Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center. It describes the research design, system architecture, use-case diagram, context-level data flow diagram, software and hardware requirements, respondents of the study, research instrument, locale of the study, data-gathering procedures, and methods of data analysis.

The methodology was structured to support both the development of the system and the evaluation of its software quality. The system development followed an Agile approach, while the evaluation was based on the ISO/IEC 25010:2011 software quality model. The procedures described in this chapter provide the basis for identifying system requirements, implementing the application, gathering evaluation data, and analyzing the results of the study.

3.1 Research Design

This study employed a developmental research design in the development and evaluation of the Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center. The developmental approach was appropriate because the study focused on identifying existing operational problems, translating identified requirements into a working information system, and evaluating the quality of the developed application.

The development of the system followed an Agile development approach. The system was developed through iterative activities that included requirements gathering, planning, design, development and coding, testing, review, implementation, and refinement. This approach allowed system functions to be

developed and improved in manageable increments based on identified operational requirements and feedback from intended users and stakeholders.

The study considered the existing processes related to animal bite case recording, vaccination monitoring, treatment follow-up, center information management, and report preparation. Information gathered from these processes served as the basis for defining the functional requirements of the system. The developed application integrated client and exposure records, vaccination scheduling and monitoring, cross-center continuation of treatment, treatment-center and vaccine information, automated email reminders, patient-accessible vaccination history, and report generation.

For software quality evaluation, the study adopted the ISO/IEC 25010:2011 software product quality model. The evaluation focused on the eight quality characteristics of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The results of the evaluation were used to determine the extent to which the developed system satisfied the identified software quality criteria.

Through this research design, the study combined the development of the Integrated Management System with a structured evaluation of its quality and suitability for the intended operational setting.

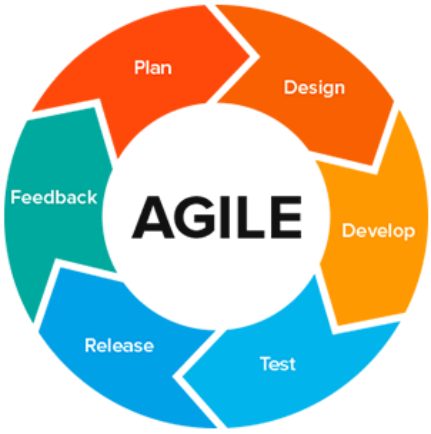


Figure 3.1. Agile SDLC model.

3.1.1 Requirements Gathering

The requirements of the system were gathered through consultation and interviews with personnel involved in animal bite treatment, vaccination monitoring, reporting, and administration. The existing procedures for client registration, animal bite assessment, vaccination scheduling, follow-up, record retrieval, center monitoring, and report preparation were examined to determine the operational problems and information requirements of the intended users.

The identified requirements served as the basis for the functions of the Integrated Management System. These included client pre-registration, animal bite case assessment, vaccination scheduling and monitoring, cross-center continuation of treatment, treatment-center and vaccine availability information, patient vaccination-history retrieval, automated email reminders, and report generation.

3.1.2 Design and Architecture

The gathered requirements were analyzed and translated into the system design. Diagrams such as the use-case diagram, context-level data flow diagram, and database structure were used to describe the interaction of users, processes, and data within the system.

The system was designed as a web-based application using a centralized architecture. The design allows participating Animal Bite Treatment Centers, Rural Health Units, and the Provincial Health Office to access appropriate system functions based on user roles and permissions. The database structure was designed to maintain relationships among client records, bite information, exposure assessments, vaccination schedules, administered doses, treatment centers, users, human rabies records, and reporting information.

3.1.3 Development and Coding

The Integrated Management System was developed as a web-based application. Vue 3 was used for the user interface, while Laravel 10 was used for

the application programming interface and server-side processing. A relational database was used to store and manage the system's records.

The system was developed incrementally following Agile principles. Functional components were implemented and refined according to the identified requirements. These included client management, online pre-registration, case assessment, immunization records, vaccination schedules, cross-center treatment continuation, vaccine availability, human rabies records, dashboard monitoring, email reminders, QR-based record retrieval, vaccination-history verification, vaccination slip generation, and report generation.

Version control and source-code management were used during development to track changes and support the continuous improvement of the application. System functions were reviewed and adjusted as operational requirements and issues were identified.

3.1.4 Quality Assurance and Software Testing

The developed system underwent functional testing to verify whether the implemented features operated according to the identified requirements. Testing covered major modules such as client registration, case assessment, immunization, vaccination scheduling, cross-center continuation of treatment, online services, email reminders, QR-based record retrieval, reporting, and administrative functions.

The system was also tested for user-access restrictions, data retrieval and updating, vaccination schedule handling, report generation, and computation of report values. Errors and inconsistencies identified during testing were corrected and retested as part of the iterative development process.

The software quality of the system will be evaluated using the ISO/IEC 25010:2011 software product quality model, particularly the eight characteristics of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

3.1.5 Implementation

The system was recently deployed to the Cagayan Provincial Health Office for initial operational testing. During this stage, PHO personnel are using the system and reviewing its functions based on their actual workflows and requirements.

The initial implementation allows authorized users to test functions related to animal bite assessment, vaccination monitoring, client records, schedule management, center information, reporting, and other operational features of the system. The Provincial Health Office is also providing observations and recommendations based on their experience during the testing period.

These recommendations are considered as part of the continuing refinement of the system before wider or full-scale implementation.

3.1.6 Maintenance and Support

The system is currently undergoing continuous refinement based on issues identified during testing and recommendations provided by the Provincial Health Office. Maintenance activities include bug fixing, adjustment of existing functions, interface improvements, report corrections, and enhancement of vaccination and scheduling processes.

System modifications are reviewed and implemented as necessary to improve usability, reliability, and alignment with operational requirements. Server-level maintenance and automated daily backup procedures also support data protection and continued system availability.

3.2 System Architecture

A multi-tier web application architecture was adopted for the development of the Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center. The architecture separates the presentation, application, and data layers to support security, maintainability, performance, and future system expansion.

A. Presentation Layer. The presentation layer consists of the web-based user interface accessed through a standard web browser. It provides interfaces for public users, health personnel, and administrators. Public functions include online pre-registration, viewing treatment-center information and vaccine availability, and vaccination-history verification. Authorized health personnel use the system for animal bite assessment, client-record retrieval, vaccination scheduling and monitoring, dose documentation, and report generation. Administrative interfaces support user accounts, treatment centers, signatories, and provincial reporting functions.

B. Application Layer. The application layer contains the core business logic and services of the system. It handles authentication and access control, client and exposure management, vaccination schedules, administered doses, QR-based record retrieval, vaccine availability, human rabies records, report processing, and system validation. Laravel Sanctum is used to protect authenticated operational routes, while additional restrictions are applied according to user roles and treatment-center ownership. The application layer also supports cross-center vaccination, allowing authorized personnel from another participating center to retrieve the patient's vaccination information and record a succeeding dose while preventing modifications to the original assessment created by another center.

C. Data Layer. The data layer provides centralized storage for client and guardian records, bite and exposure information, vaccination schedules and histories, treatment-center information, vaccine availability, human rabies records, user accounts, report information, and transaction logs. Centralized data storage allows authorized personnel to retrieve relevant patient and vaccination information across participating treatment centers while maintaining the relationship between records and the centers responsible for the recorded services.

The system also includes an automated email notification service for vaccination reminders. Scheduled vaccination information is processed by the IMS and sent

through the configured mail service, which uses Microsoft Graph for email delivery. The mail service is reused from an existing project to avoid redundant configuration of email credentials and delivery settings. Patients with a vaccination scheduled for the following day receive an email reminder at approximately 6:00 PM together with the relevant schedule information and QR code.

The system also supports report generation in PDF and spreadsheet formats for center-level and provincial reporting. Network security mechanisms, including firewall protection and HTTPS/TLS, support the protection of information exchanged between users and the application.

Overall, the adopted architecture provides centralized information management while separating the user interface, application processing, data storage, and integrated supporting services of the system.

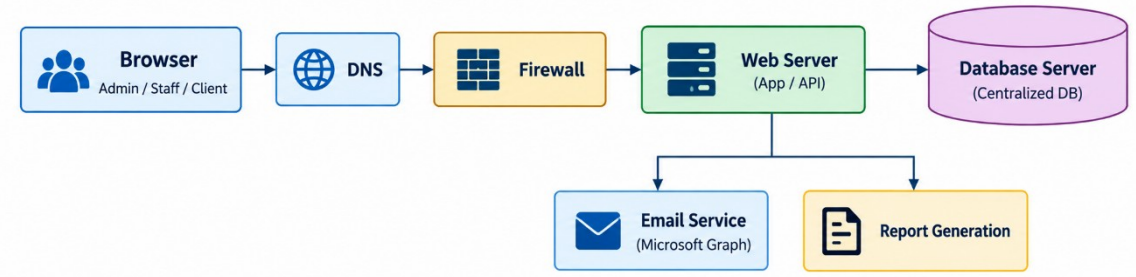


Figure 3.2. Architecture of the Integrated Management System.

3.3 Use-Case Diagram

The Use-Case Diagram illustrates the major functionalities of the Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center and identifies the key actors who interact with the system. These actors include the Administrator (PHO/PLGU Admin), Users from PLGU/MLGU/RHU, and Clients/Patients.

The diagram presents the main system functions available to each actor. Public services include online pre-registration, viewing treatment centers and vaccine

availability, and verification of vaccination history. Clients/Patients are limited to these public-facing services and do not have access to staff or administrative functions.

Authorized Users from PLGU/MLGU/RHU can view the dashboard and monitoring information, search and retrieve client records, encode client information, assess animal bite cases, record human rabies cases, create and update vaccination schedules, record administered doses, continue vaccination across participating centers, generate QR codes and vaccination slips, retrieve vaccination schedules and histories through QR-based record access, and generate center-level reports.

The Administrator has access to the same operational functions available to Users and performs additional administrative and provincial-level functions. These include managing user accounts and password resets, maintaining treatment centers, signatories, and application preferences, viewing dashboard and monitoring information, and generating provincial or Department of Health reports.

The diagram also reflects the system's role-based and center-based access restrictions. Authorized users can retrieve shared client and vaccination information needed for continuity of treatment, while modification of records is limited according to user role and treatment-center ownership.

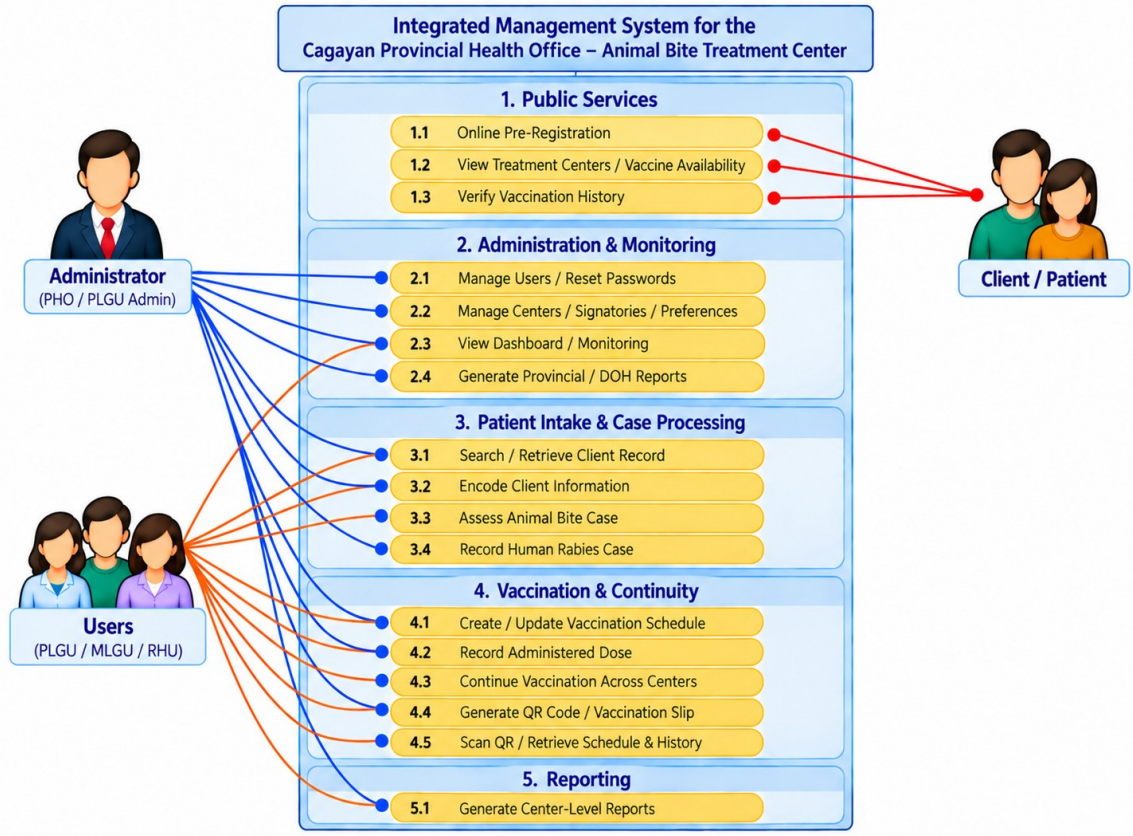


Figure 3.3. Use Case Diagram

3.4 Context Level - Data Flow Diagram

The Context Level Data Flow Diagram (DFD) presents the overall interaction of the external entities with the Integrated Management System. The external entities include the Client/Patient, Users from PLGU/MLGU/RHU, and the Administrator (PHO/PLGU Admin).

The Client/Patient submits pre-registration information and vaccination-history verification requests. In return, the system provides treatment-center information, vaccine availability information, vaccination schedules and history, and QR code or vaccination-slip information.

The Users from PLGU/MLGU/RHU submit client information, animal bite case details, vaccination updates, human rabies case details, and report requests. The system provides retrieved client records, vaccination schedules and histories, QR code or vaccination-slip information, center-level reports, and dashboard or monitoring information.

The Administrator manages user account data, treatment-center information, signatories, system settings or preferences, and report requests. In return, the system provides user account information, dashboard and monitoring data, and provincial or Department of Health reports.

The diagram shows the major flow of information between the system and its external users and illustrates how the Integrated Management System supports client services, animal bite case management, vaccination continuity, monitoring, and reporting.

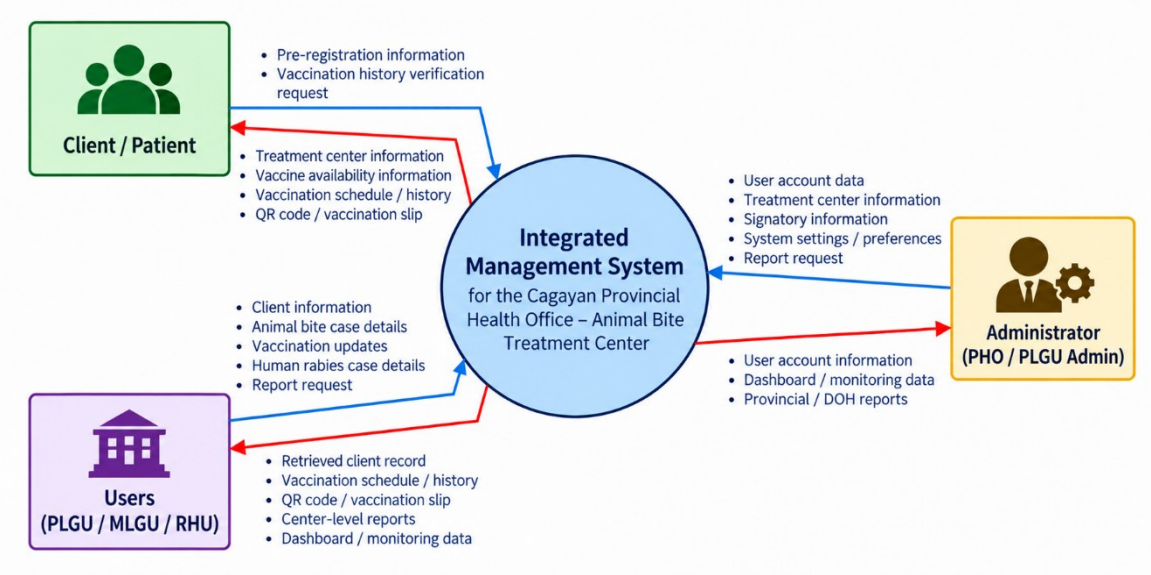


Figure 3.4. Context-level data flow diagram

3.5 Software and Hardware Requirements

Table 5. Software requirements and identified dependencies

Purpose	Implementation /requirement	Component
Build and serve browser assets	Vue 3; TypeScript 5.9; Vite 7; Node.js compatible with installed Vite	Frontend

Purpose	Implementation /requirement	Component
Navigation, state, requests and QR handling	Vue Router; Pinia; Tailwind CSS; Axios; QR/ZXing libraries	Interface libraries
API and business operations	Laravel 10; PHP constraint ^8.1; Composer dependencies	Backend
Persistent records and queries	MySQL-compatible relational server; existing schema and views	Database
Authenticated requests and session checks	Laravel Sanctum and application middleware	Authentication
PDF and Excel generation	Snappy / compatible wkhtmltopdf binary; PhpSpreadsheet	Report output
Data-change events	Laravel Reverb / Echo when configured	Realtime
Process due reminder requests	Scheduler plus configured HTTP email endpoint	Reminder delivery
Staff and public application access	Current Chrome or another tested modern browser	Browser

Table 6. Suggested hardware baseline for pilot planning

Qualification	Suggested baseline	Environment
Planning recommendation, not benchmarked minimum	Modern dual-core CPU; 8 GB RAM; 1366 x 768 or larger display	Staff workstation
Test permission and scanning conditions	Camera or QR-capable scanner where the workflow requires it	Camera / scanner
Size after measuring workload and data growth	4 CPU cores; 8 GB RAM; SSD storage with backup capacity	Pilot server
No verified offline synchronization	Stable LAN or internet connection; protected server access	Network
Recovery must be tested before reliance	Backup storage and appropriate power protection	Continuity

These hardware figures are planning recommendations rather than measured limits. The application repositories establish software dependencies but do not establish maximum user capacity. Before deployment, administrators should measure response times and resource use under representative concurrent activity and confirm compatibility of all installed packages and report binaries.

3.6 Respondent of the Study

The respondents of the study were purposively selected personnel from the Cagayan Provincial Health Office who were directly involved in the evaluation of the Integrated Management System. A total of 20 respondents participated, consisting of 10 IT experts, one doctor, three nurses, one midwife, one nursing attendant, and four administrative staff.

The respondents were selected based on their roles and relevance to the operation, supervision, administration, and technical evaluation of the system. The Doctor/Head of PHO represented the management and supervisory perspective; the nurses, midwife, and nursing attendant evaluated workflows related to client handling, assessment, vaccination monitoring, and treatment documentation; the administrative staff evaluated records, reporting, and office-related functions; and the IT experts evaluated the technical and software quality characteristics of the developed system.

Although the system is intended for use by other participating treatment centers in the province, personnel from those centers were not included as respondents in the present evaluation.

Position/Role	Frequency (f)	Percentage (%)
IT Expert	10	50.00%
Doctor	1	5.00%
Nurse	3	15.00%
Midwife	1	5.00%
Nursing Attendant	1	5.00%
Administrative Staff	4	20.00%
TOTAL	**20**	**100%**

Figure 3.5. Position/Role of the Respondents

The respondents of the study were selected based on their actual participation in the evaluation and their relevance to the operation, administration, or technical assessment of the Integrated Management System. All respondents were personnel from the Cagayan Provincial Health Office who were formally included in the evaluation process. Database accounts, registered treatment centers, and public users of the system were not automatically considered study participants unless they were specifically recruited and included in the evaluation.

3.7 Research Instrument

The study used researcher-developed instruments to gather information regarding the existing animal bite treatment processes and to evaluate the developed Integrated Management System.

A semi-structured interview questionnaire was used to identify the problems and challenges encountered in animal bite case recording, vaccination monitoring, record retrieval, and report preparation. The interview questions also gathered information regarding the existing workflow and the requirements of health and administrative personnel. The responses served as a basis for determining the operational requirements of the system.

For the evaluation of the developed system, an ISO/IEC 25010:2011-aligned questionnaire was used. The questionnaire was based on the eight software product quality characteristics of functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. The instrument was researcher-developed and adapted to the functions and intended users of the Integrated Management System.

The questionnaire used a five-point rating scale to determine the respondents' assessment of the software quality of the developed system. Respondents evaluated the characteristics and system functions relevant to their knowledge and experience. The IT experts primarily contributed to the technical evaluation, while the PHO health and administrative personnel evaluated the system based on its functionality, usability, performance, reliability, and relevance to their operational activities.

The research instruments were reviewed to ensure that the questions were understandable and aligned with the objectives of the study. No patient-identifying information was included in the evaluation questionnaire.

3.8 Locale of the Study

The study was conducted at the Cagayan Provincial Health Office in the Province of Cagayan, Philippines. The Provincial Health Office served as the primary locale for the initial deployment, testing, and evaluation of the Integrated Management System.

The system was developed to support the management of animal bite and rabies-related information within the Provincial Health Office and to extend its use to participating Animal Bite Treatment Centers, Municipal Local Government Units, and Rural Health Units throughout the province. Following initial PHO testing and evaluation, the Provincial Health Office intends to require participating centers to use the system for recording and maintaining relevant animal bite treatment information and for supporting provincial monitoring and reporting.

The respondents included in the present study were limited to personnel from the Provincial Health Office.

3.9 Data Gathering

The researchers followed the following procedures in gathering the data needed for the study:

1. Securing permission. Permission was obtained from the concerned office to conduct the study, gather information regarding the existing animal bite treatment processes, and evaluate the developed system.
2. Review of existing processes and documents. Existing forms, reporting requirements, procedures, and records related to animal bite case management, vaccination monitoring, record retrieval, and report preparation were reviewed to identify the current workflow and information requirements.
3. Conduct of interviews. Selected personnel from the Provincial Health Office were interviewed using a semi-structured interview guide. The interview focused on the problems encountered in the existing process, workflow requirements, reporting needs, and recommendations for the development of the system.

4. Identification of system requirements. Information gathered from interviews and document review was organized and used as the basis for determining the functional and operational requirements of the Integrated Management System.
5. Development and testing of the system. The identified requirements were implemented through the Agile development approach. System functions were developed, tested, reviewed, and refined throughout the development process.
6. Initial deployment and orientation. The developed system was deployed to the Cagayan Provincial Health Office for initial operational testing. PHO personnel were oriented on the major functions and workflows of the system and were given the opportunity to use and review the application.
7. System evaluation. The selected respondents evaluated the developed system using the ISO/IEC 25010:2011-aligned questionnaire. A total of 20 respondents participated in the evaluation, consisting of 10 IT experts, one doctor, three nurses, one midwife, one nursing attendant, and four administrative staff.
8. Collection of observations and recommendations. Comments, observations, and recommendations provided by PHO personnel during testing and evaluation were documented and considered for further refinement of the system.
9. Data consolidation. Completed questionnaires and interview responses were checked for completeness, organized, tabulated, and prepared for analysis.

3.10 Data Analysis

The data gathered through the preceding section were tallied and organized for proper analysis and interpretation. Frequency Count and Mean were used to interpret the respondents' assessment, comments, and recommendations regarding the developed Integrated Management System.

The Mean was also used to determine the extent of software quality of the system based on the eight characteristics of ISO/IEC 25010:2011.

In determining the efficiency of the Integrated Management System compared with the existing manual process, the researcher adapted the formula used in the reference study:

$$\text{Efficiency of the IMS} = \frac{\text{Manual Time} - \text{IMS Time}}{\text{IMS Time}} \times 100\%$$

The researcher measured the processing time required to complete equivalent tasks using both the existing manual process and the developed Integrated Management System. The comparison was used to determine the difference in processing time between the two methods.

A five-point Likert Scale was utilized to obtain the point scales and descriptive equivalents of the respondents' responses. The scale was used in the assessment of the developed system and in determining the extent of software quality based on ISO/IEC 25010:2011.

Table 8. Likert Scale for the Assessment of the Integrated Management System

Scale point	Interpretation	Range
5	Strongly Agree	4.21-5.00
4	Agree	3.41-4.20
3	Neither Agree nor Disagree	2.61-3.40
2	Disagree	1.81-2.60
1	Strongly Disagree	1.00-1.80

Table 9. Likert Scale for the Extent of Software Quality Based on ISO/IEC 25010:2011

Scale point	Interpretation	Range
5	Very high extent	4.21-5.00

Scale point	Interpretation	Range
4	High extent	3.41-4.20
3	Moderate extent	2.61-3.40
2	Low extent	1.81-2.60
1	Very low extent	1.00-1.80

CHAPTER IV

RESULTS AND DISCUSSIONS

The chapter represents the analysis and interpretation of data gathered across the Integrated Management System for Cagayan Provincial Health Office – Animal Bite Treatment Center.

4.1 Problems Encountered in the Existing System

This section presents the problems identified in the existing process of animal bite case recording, vaccination monitoring, record retrieval, and report preparation. The findings were based on the interviews, observations and review of the existing workflow conducted during the requirements-gathering stage of the study.

The identified problems included the following:

a. Repeated encoding of client and treatment information

b. Difficulty in retrieving previous vaccination records
c. Challenges in monitoring scheduled and overdue vaccination doses
d. Difficulty in consolidating animal bite and rabies-related information for reporting.

These issues were particularly important because a patient may receive treatment over several visits and may continue succeeding vaccination doses at another participating treatment center.

Also, the preparation of provincial reports is a problem. Animal bite, vaccination, biting-animal, and human-rabies information must be organized according to the required reporting structure. When information is maintained through separate records or prepared manually, additional time and effort are needed to verify, consolidate, and summarize the data.

The existing process also presented limitations in providing patients with immediate access to treatment-center information, vaccine availability, vaccination history, and reminders for upcoming doses. These concerns served as the basis for defining the functional requirements of the developed system.

4.2 The Developed System

The developed Integrated Management System provides a centralized platform for managing animal bite treatment information and supporting provincial monitoring and reporting. It integrates client and guardian records, animal bite and exposure information, vaccination schedules, administered doses, treatment-center information, vaccine availability, human rabies records, user accounts, and report-related information.

The system provides public services for client pre-registration, viewing treatment-center information and vaccine availability, and verification of vaccination history. Authorized users can retrieve client records, record animal bite assessments, manage vaccination schedules, document administered doses,

continue vaccination across participating centers, generate vaccination slips with QR codes, record human rabies cases, and generate center-level reports.

The Administrator has access to the operational functions available to regular users and additional functions for managing user accounts, password-reset requests, treatment centers, signatories, system preferences, dashboard monitoring, and provincial or Department of Health reports.

The system also includes automated email reminders for patients with scheduled vaccinations. Reminder information is processed by the system and sent through the configured email service one day before the scheduled vaccination. The email includes relevant schedule information and the QR code associated with the patient's exposure record.

4.2.1 Hierarchical Input Process Output (HIPO)

The Hierarchical Input-Process-Output diagram presents the major system modules and their relationship within the Integrated Management System. It shows how the system is organized into functional areas for public services, user operations, administrative functions, vaccination management, monitoring, and reporting.

The HIPO structure illustrates the hierarchy of the system beginning with the main application and proceeding to its major functions and subfunctions. It provides a summarized representation of how the different modules of the system work together to support client services, animal bite case management, vaccination monitoring, administrative functions, and provincial reporting.

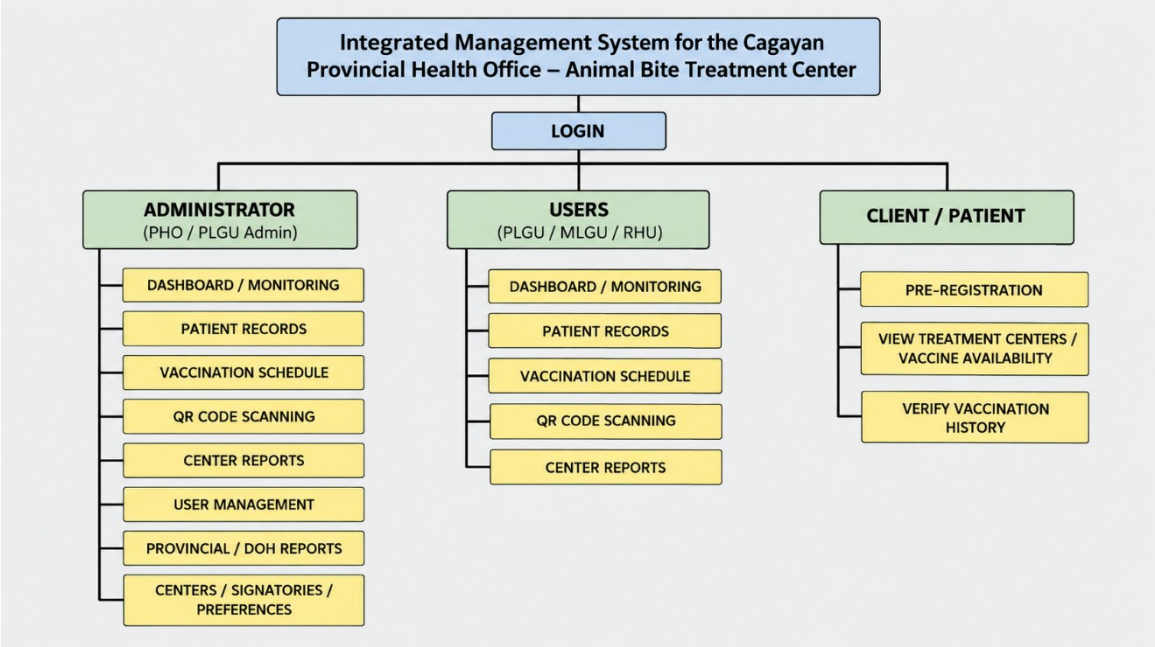


Figure 4.1. HIPO diagram of IMS

4.2.2 Input Process Output (IPO) Diagram

The Input-Process-Output diagram presents the flow of information through the developed system. The inputs consist of client and guardian information, animal bite and exposure details, vaccination and treatment information, treatment-center data, human rabies information, user information, and reporting requirements.

The processes include client registration and retrieval, animal bite assessment, creation and updating of vaccination schedules, recording of administered doses, cross-center continuation of treatment, generation of QR-based vaccination records and vaccination slips, monitoring of vaccination schedules, management of vaccine availability, recording of human rabies cases, automated email reminder processing, and report generation.

The outputs include centralized client and exposure records, updated vaccination histories, monitored vaccination schedules, QR-based record retrieval, printable vaccination slips, automated vaccination reminders, treatment-center and vaccine availability information, dashboard and monitoring data, center-level accomplishment reports, and consolidated provincial and Department of Health reports.

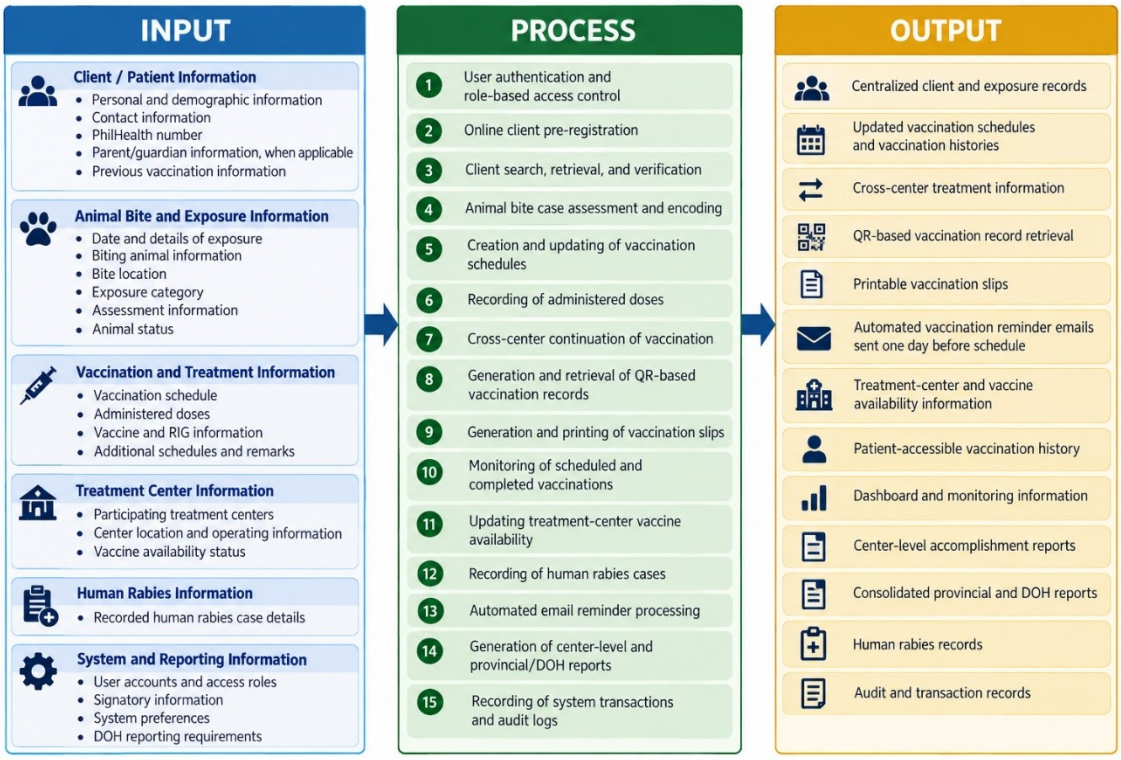


Figure 4.3. IPO diagram of the IMS

4.2.3 Entity Relationship Diagram (ERD)

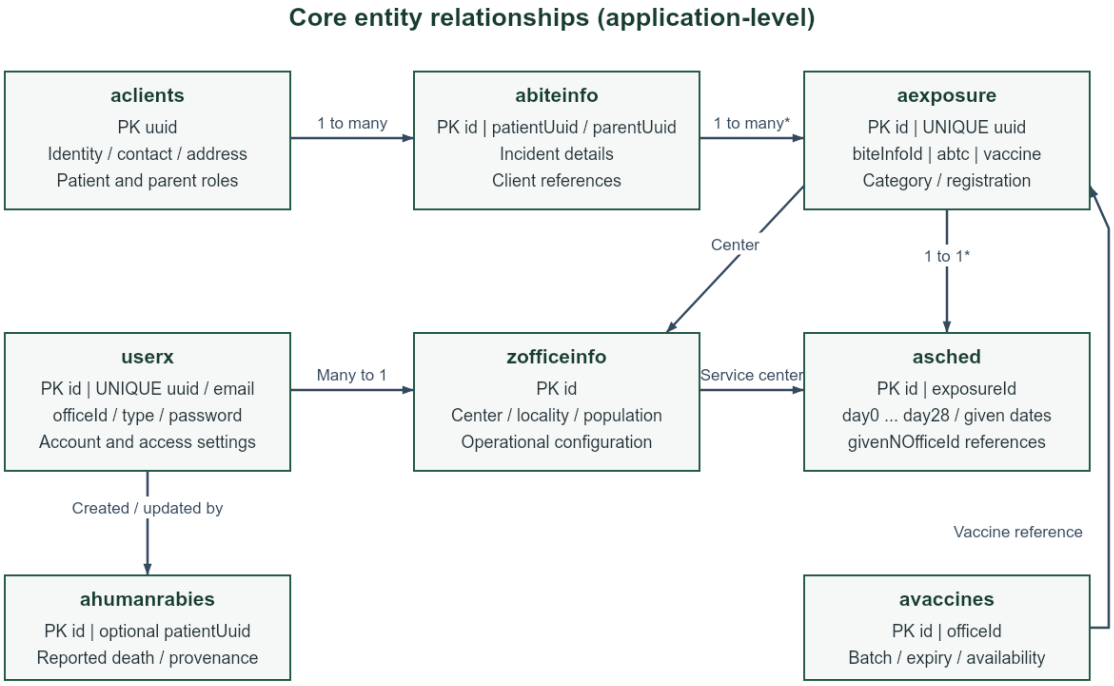


Figure 4.5. Core data relationships

The diagram summarizes application relationships inferred from controller joins and confirmed column names. The asterisk denotes workflow cardinality that must not be mistaken for a database-enforced uniqueness constraint. Additional locality, audit, signatory, and supporting authentication objects are documented in the table design.

The client table uses a UUID primary key. Bite information stores the patient UUID and an optional parent UUID. Exposure records identify the related bite information, center, registration sequence, and vaccination-related selections. Schedules identify the exposure and keep separate planned and administered date fields. Vaccine and center tables supply references used by these operational records.

The current schedule structure stores named dose-day columns in one row. This matches the implemented interface but is less flexible than a child table

containing one row per dose. A future normalization change would require migration of existing records and careful updates to reports, reminders, and retrieval logic. The ERD therefore documents the current structure rather than silently replacing it with an idealized schema.

4.2.3.1 User's View

The following figures are screenshots of the running application, not recreated interface mockups. They show the local demonstration state, and their numerical values must not be interpreted as verified province-wide health statistics. Empty forms and aggregate views were selected to explain the interface without publishing patient histories. Screenshots establish visible layout and controls; they do not prove every action has passed testing.

Sign-in view

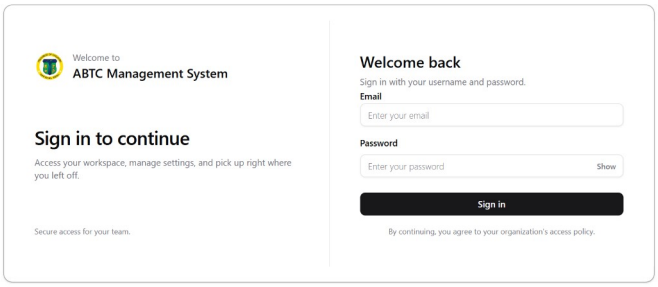


Figure 4.6. Sign-in view: actual running application

Purpose: accepts staff credentials and provides entry to the authenticated workspace.

The user enters an assigned email address and password. Successful authentication allows the route guard to retrieve the profile and establish the account context. Invalid or expired credentials should return the user to the sign-in workflow. Public online services have a separate route.

Dashboard view

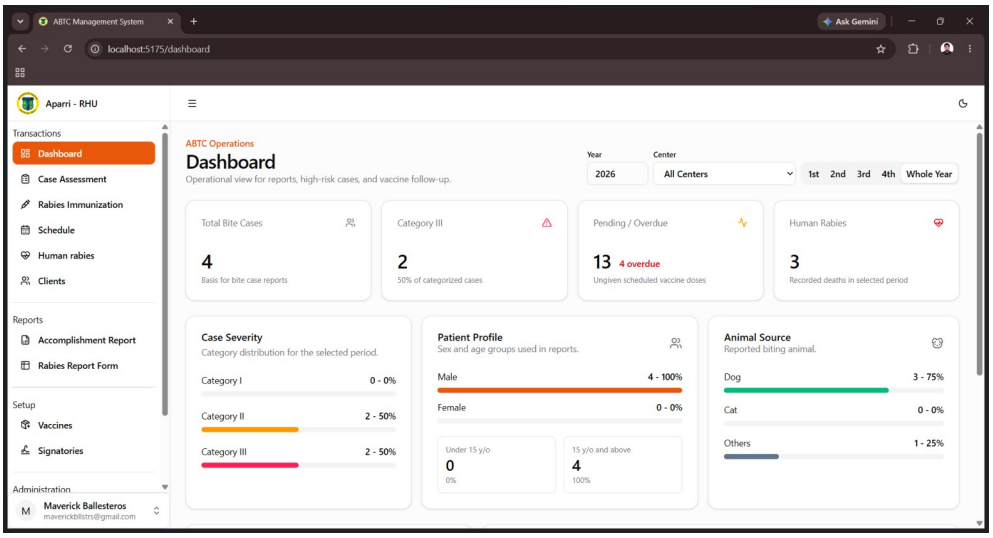


Figure 4.7. Dashboard view: actual running application

Purpose: presents the selected period's operational summaries and categories in one location.

The user selects a year, center where available, and quarter or full-year period. Cards distinguish cases, higher-category exposures, pending or overdue doses, and recorded human rabies. The period and population represented by each measure must be understood before comparisons are made.

Case assessment view

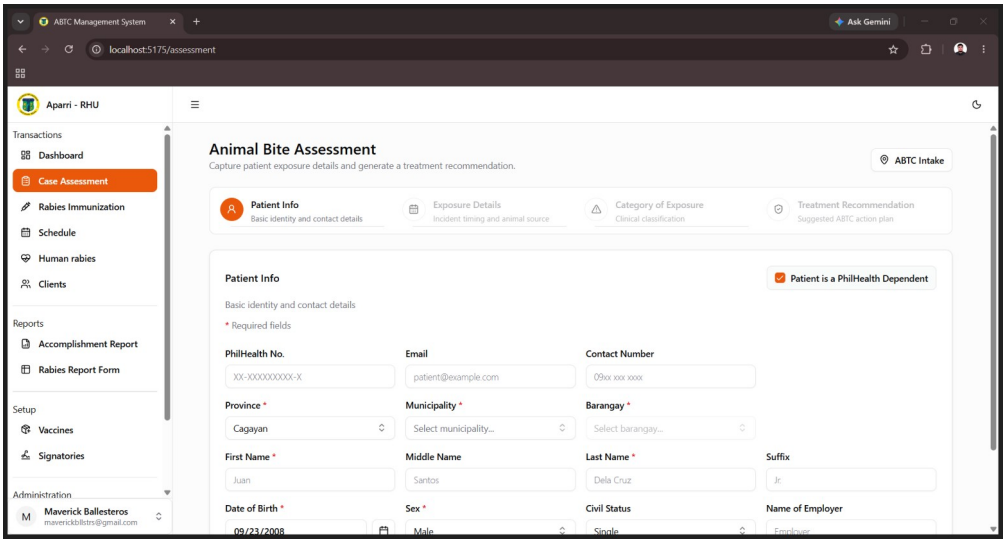


Figure 4.8. Case assessment view: actual running application

Purpose: starts a structured workflow for client identity, incident details, exposure category, and treatment-related recording.

The displayed first step captures identity and contact information and the dependent relationship when relevant. Later steps collect the clinical record. Staff must verify the person and the selected episode before saving. The interface contains treatment recommendation wording; this should be governed by professional review and should not be presented as an independently validated diagnostic capability.

Rabies immunization view

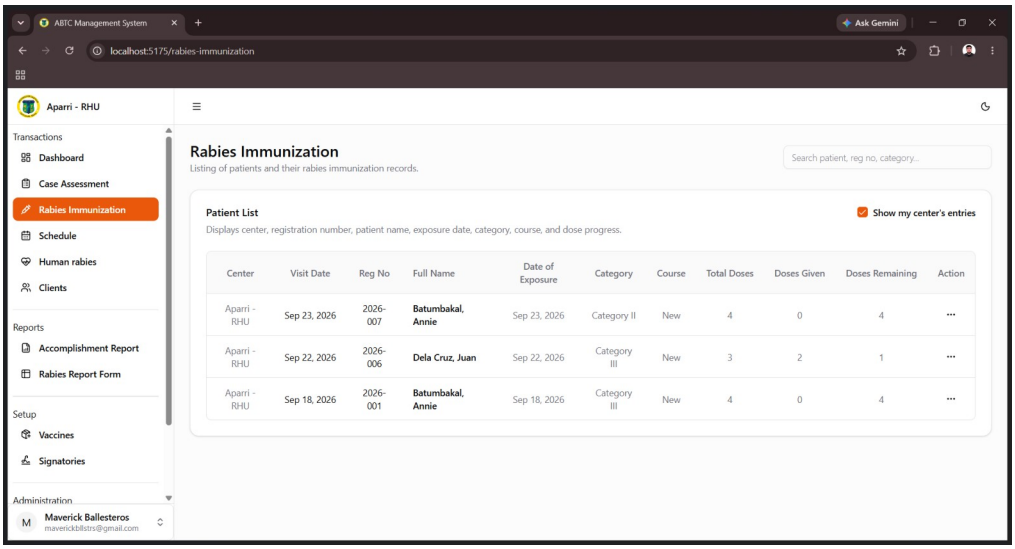


Figure 4.9. Rabies immunization view: actual running application

Purpose: lists patient exposure episodes and dose progress for review and follow-up.

The user searches by patient, registration number, or category and confirms the correct center and episode before opening an action. The list distinguishes total, given, and remaining doses; these values should be reconciled with the recorded administration dates.

Vaccination schedule view

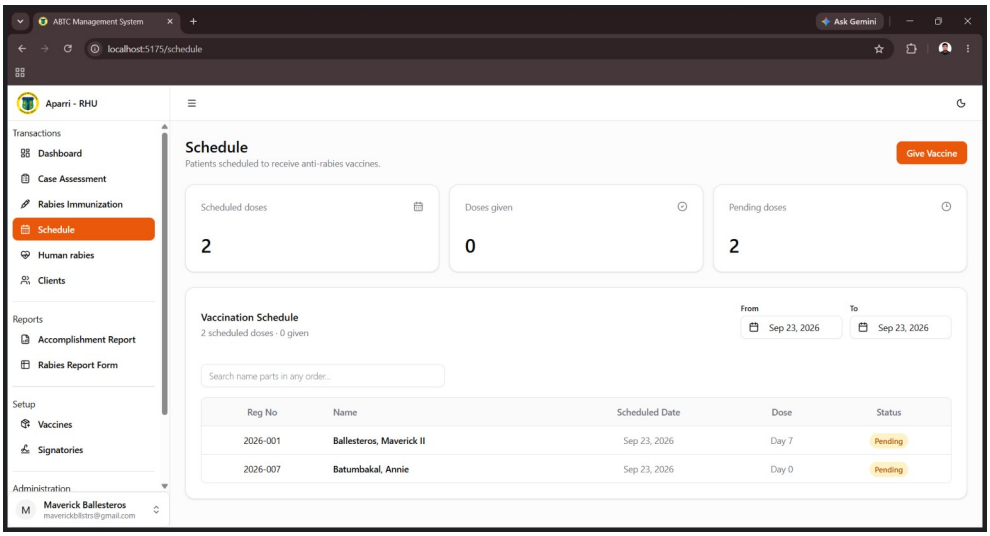


Figure 4.10. Vaccination schedule view: actual running application

Purpose: presents scheduled, completed, and pending doses for a selected date interval.

The user selects a date range, searches for the intended patient, verifies the registration and scheduled dose, and then uses the authorized Give Vaccine workflow. A planned date must remain distinct from the date on which a dose was actually administered.

Client records view

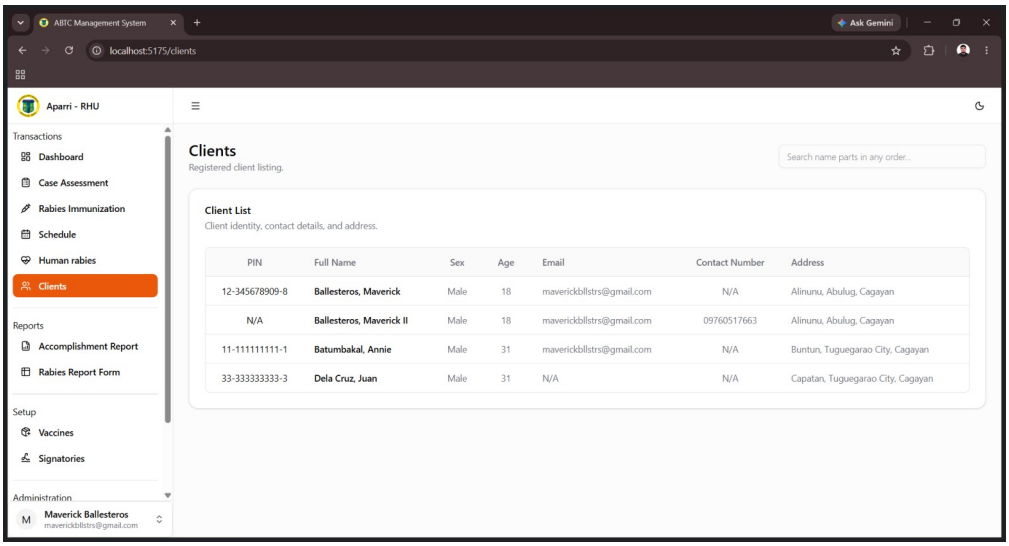


Figure 4.11. Client records view: actual running application

Purpose: provides a searchable directory of registered client identity and contact information.

The user compares more than one identifier before selecting a person. Duplicate or incorrect identity records should be referred through the authorized correction process because they may already be linked to incidents, exposures, and schedules.

Human rabies surveillance view

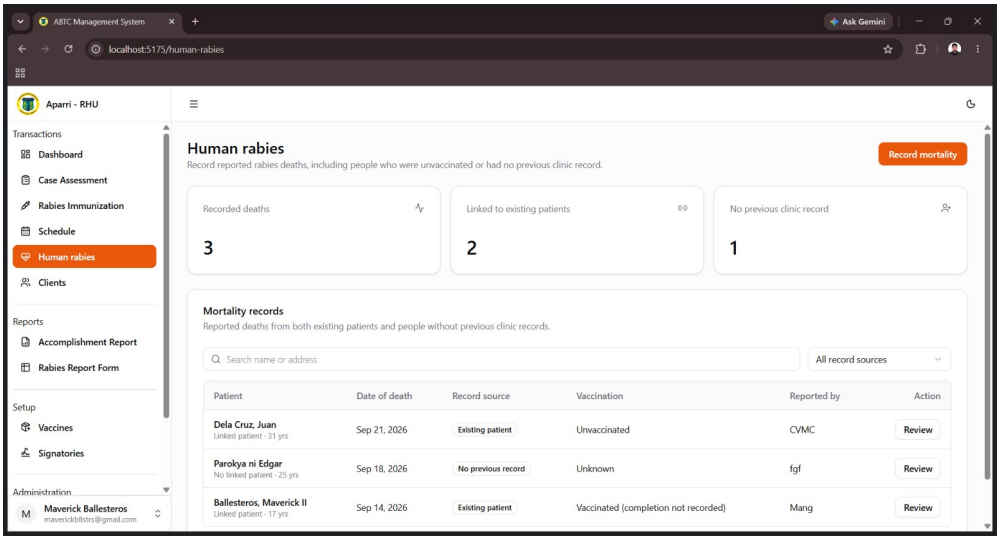


Figure 4.12. Human rabies surveillance view: actual running application

Purpose: records reported human rabies deaths and distinguishes records linked to existing clients from reports without a previous clinic record.

Authorized staff review source, vaccination status, event date, and reporting provenance. A displayed entry represents a report recorded in the system and must not be described as laboratory confirmation unless the supporting evidence establishes that status.

Public online services view

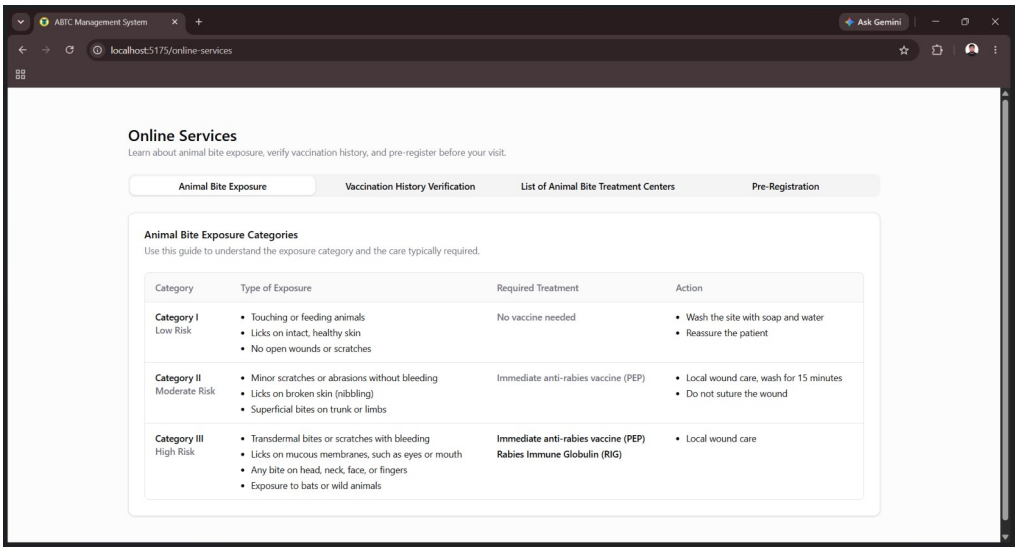


Figure 4.13. Public online services view: actual running application

Purpose: groups the publicly available information, verification, center directory, and pre-registration functions.

The visible tab presents exposure-related information; the other tabs lead to history verification, center details, and intake. The clinical content shown is captured as part of the application and requires review against approved guidance. In particular, wording about exposure categories should be reviewed before public reliance. The screenshot is not a clinical reference.

Accomplishment report view

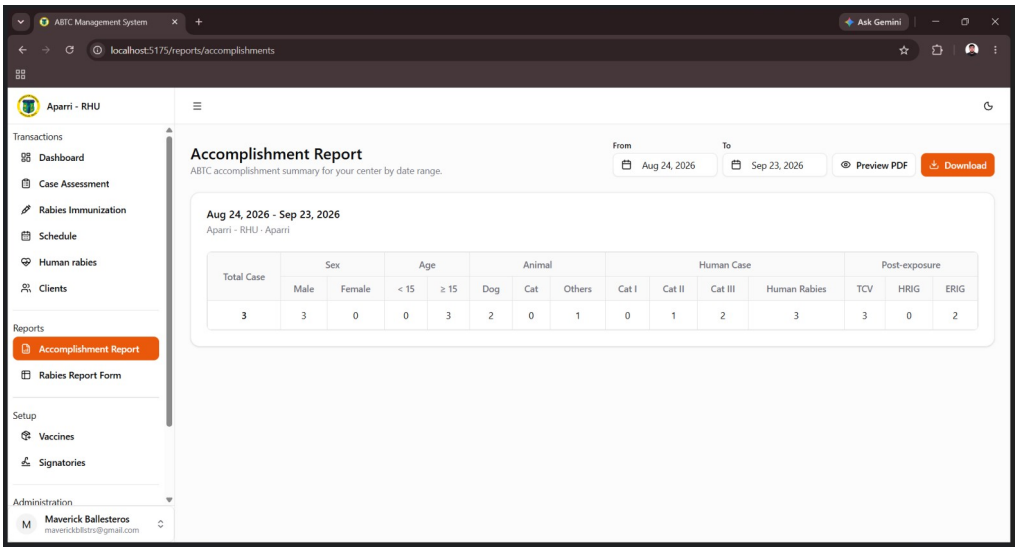


Figure 4.14. Accomplishment report view: actual running application

Purpose: summarizes activity for a selected date range and the applicable center context.

The user chooses the start and end dates and reviews the summary before using Preview PDF or Download. Sex, age group, source animal, exposure category, human rabies, and post-exposure fields appear in the table. Report reconciliation should verify the exact date basis and count definition used by each column.

Bite cases report view

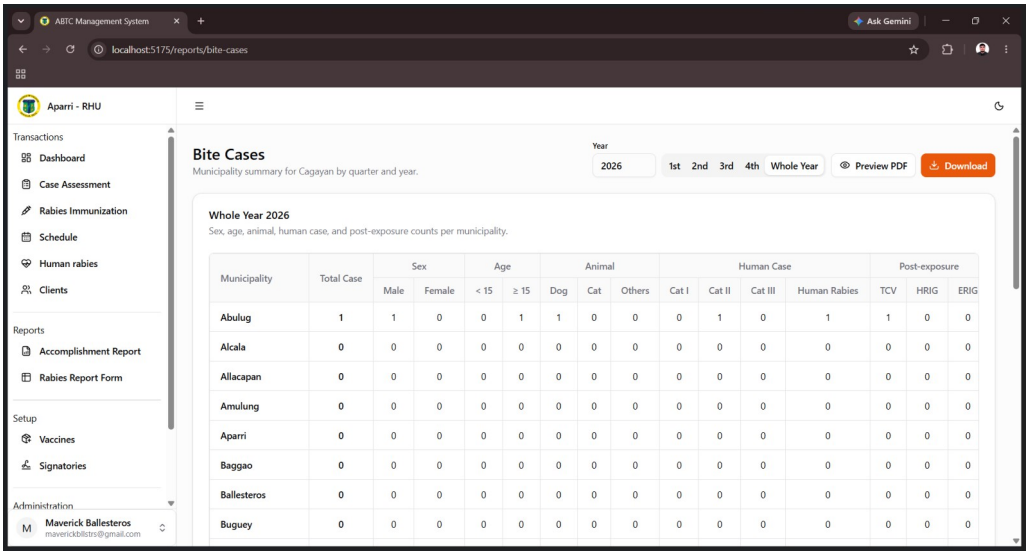


Figure 4.15. Bite cases report view: actual running application

Purpose: displays municipality-level summaries for an administrator-selected year and reporting period.

The quarter buttons and whole-year option change the period. PDF actions provide a reporting output path. Because this view is broader than an individual center report, the application checks administrator access. A direct unauthorized API request should be included in the formal verification plan.

Cohort report view

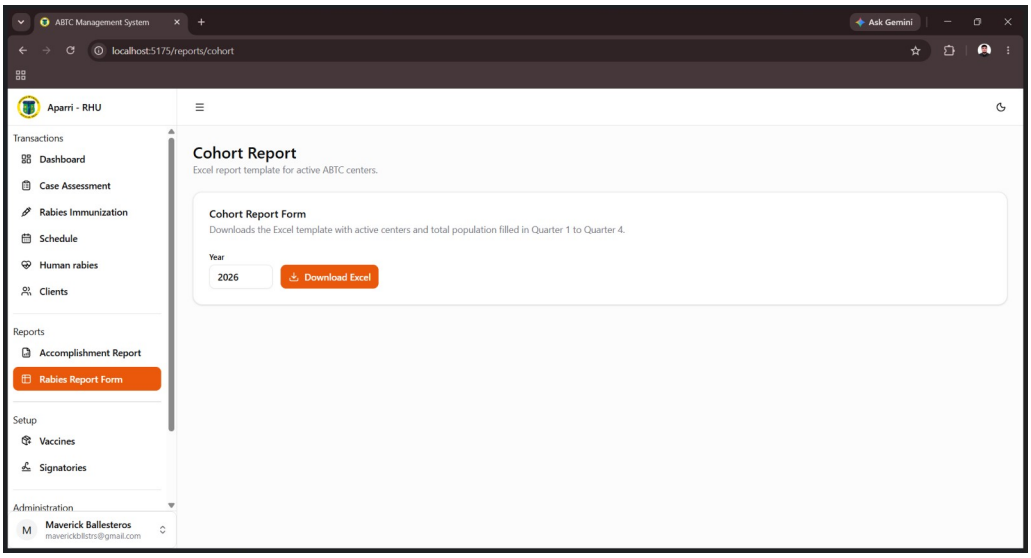


Figure 4.16. Cohort report view: actual running application

Purpose: provides the year selection and Excel output for the rabies reporting template.

The interface states that active centers and population information are populated across the template's quarters. This function should be described as template preparation unless every clinical indicator is confirmed to be automatically calculated. Downloaded workbooks require review of headings, year, center list, and populated values.

Vaccine setup view

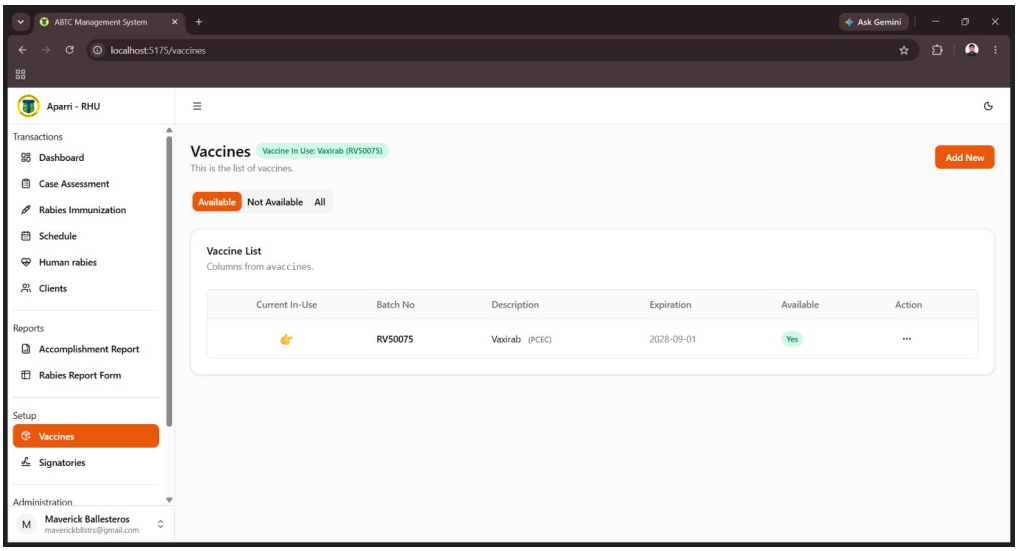


Figure 4.17. Vaccine setup view: actual running application

Purpose: maintains vaccine batch descriptors, expiry, availability, and the current in-use selection.

The user reviews the available, unavailable, or complete list and uses permitted actions to maintain the record. An availability flag is a service indicator and should not be interpreted as a complete stock balance. Changes should reflect an authorized staff confirmation.

Treatment-center administration view

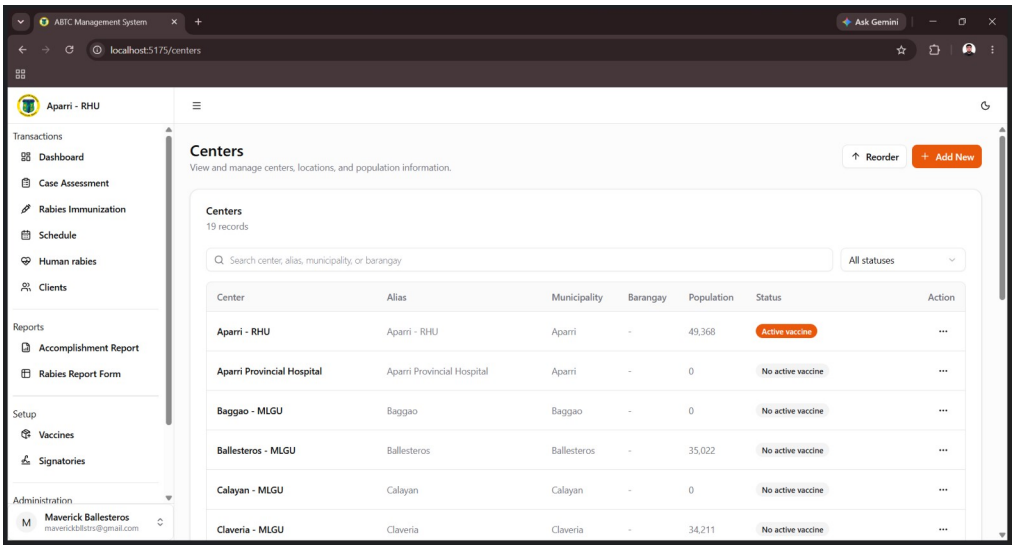


Figure 4.18. Treatment-center administration view: actual running application

Purpose: maintains center identifiers, locality information, population values, and service status.

Search and status filtering support location of a center record. Add, update, and ordering functions affect reference information used elsewhere in the application. Center population figures displayed here are configuration data, not independently validated demographic estimates.

Account administration view

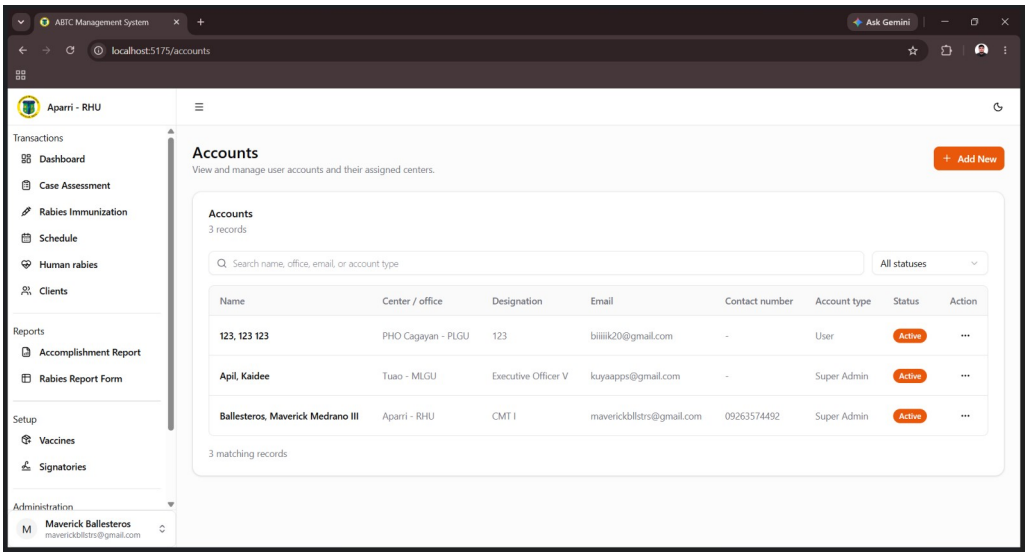


Figure 4.19. Account administration view: actual running application

Purpose: allows authorized administrators to maintain user identities, center assignments, roles, and active status.

The administrator verifies the person, assigned office, designation, account type, and status before making a change. Password generation, reset, impersonation, and privilege changes require institutional controls and review.

Report signatories view

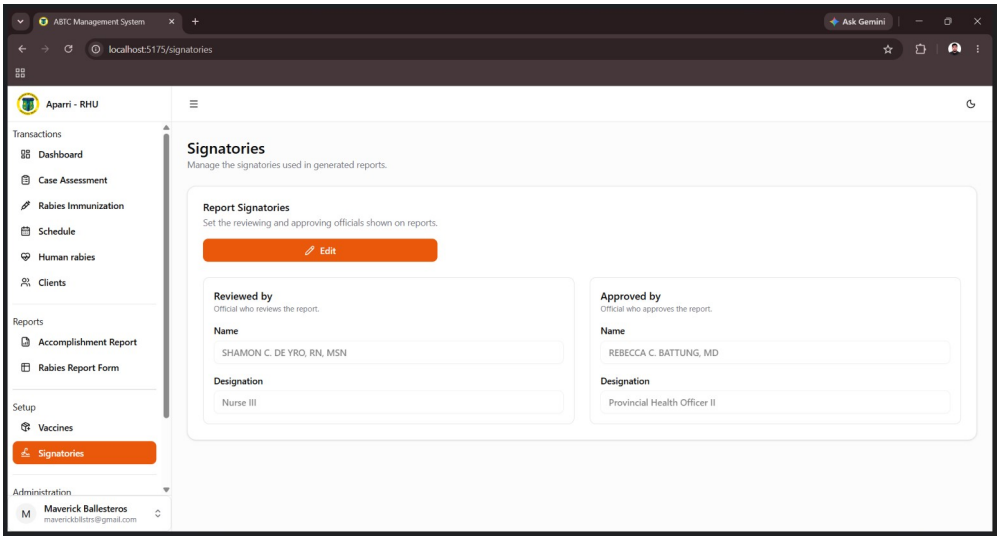


Figure 4.20. Report signatories view: actual running application

Purpose: maintains the reviewing and approving officials displayed on generated reports.

Authorized staff enter the names and designations used in report presentation and verify them through a report preview. Changes affect document accountability and should follow the office's approval process.

Personal settings view

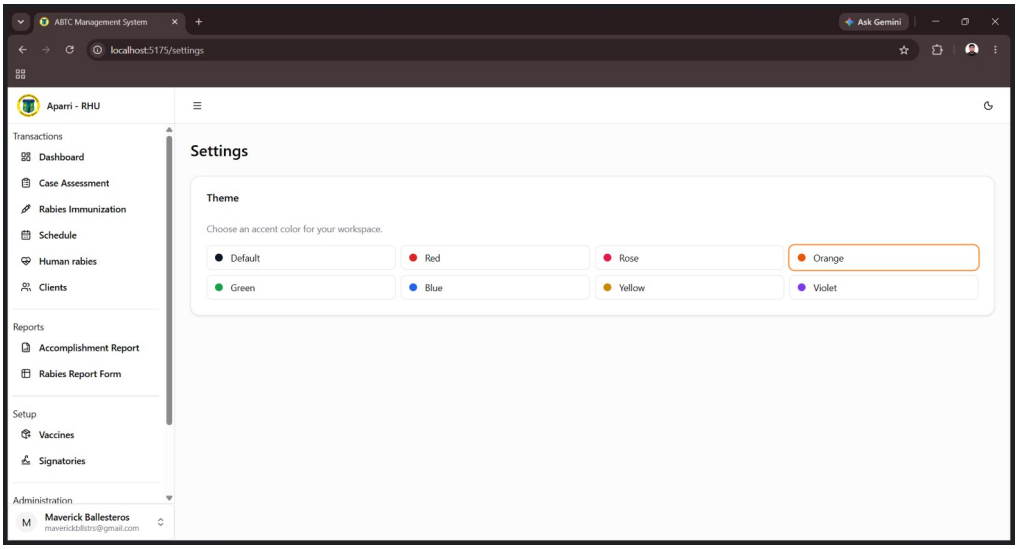


Figure 4.21. Personal settings view: actual running application

Purpose: provides account-level workspace presentation preferences.

The user selects an available theme preference for the current workspace. This setting changes presentation and does not alter clinical records or reporting data.

4.2.3.2 Data Model

Table 11. Logical data domains

Relationship / purpose	Principal objects	Domain
Reusable patient and parent identity	aclients; locality reference tables	Identity
Incident linked to assessed exposure	abiteinfo; aexposure	Incident and assessment

Relationship / purpose	Principal objects	Domain
Planned and administered dates; selected vaccine	asched; avaccines	Vaccination
Center configuration and assigned accounts	zofficeinfo; userx	Organization
Reported human rabies record with provenance	ahumanrabies	Surveillance
Change records and report signatory configuration	bauditlogs; zsignatories	Accountability
Queries over operational data	vautoemailnotif and reporting views	Derived information

The principal identity relationship is aclients.uuid to abiteinfo.patientUuid. The optional parentUuid also refers to aclients. The incident-to-assessment relationship is abiteinfo.id to aexposure.biteInfoId; the assessment-to-schedule relationship is aexposure.id to asched.exposureId. Center references include aexposure.abtc, userx.officeId, avaccines.officeId, and the given-day office fields in asched.

Database metadata shows primary, unique, and ordinary indexed columns. A column name ending in Id does not itself prove that a foreign-key constraint exists. The application relationships documented here should be reconciled with database constraints before a production migration. Duplicate prevention and data correction should be tested for both ordinary use and concurrent requests.

4.2.4 Table Design

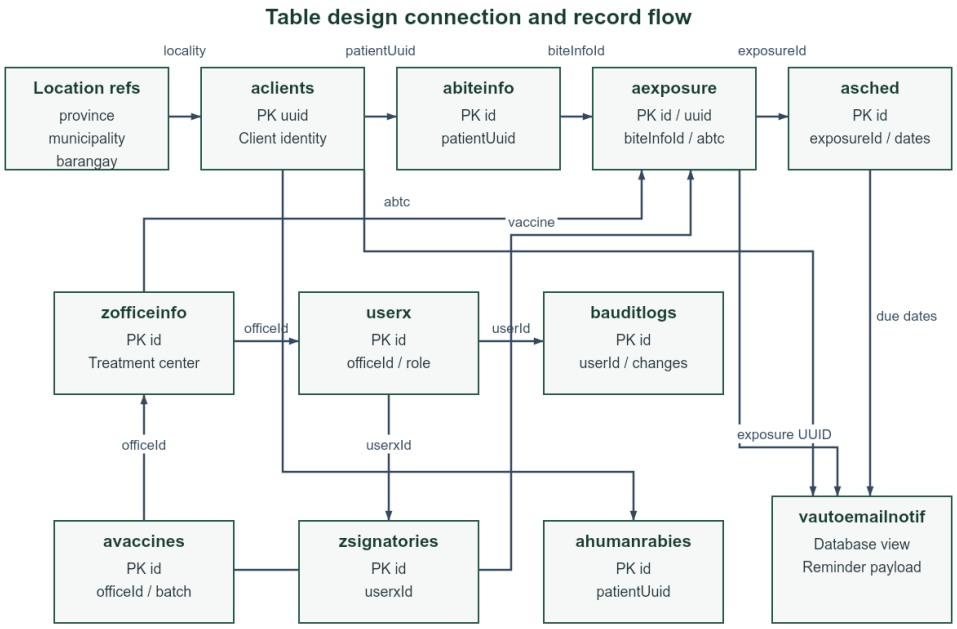


Figure 4.22. Table design connection and record flow

The simplified flow follows the principal implemented record path from locality and client identity through incident, exposure, and schedule records. Supporting center, account, vaccine, signatory, audit, surveillance, and notification objects connect to that path without implying constraints that were not observed in metadata.

The diagram is a logical connection view based on inspected column names and application joins. Arrow direction indicates reference or information flow for documentation purposes; it does not, by itself, assert that every link is enforced by a database foreign-key constraint.

Table 12. Table relationship and flow matrix

Purpose	Related table	Connection field	Source table
Limits municipalities to a province	ymunicipalities	provinceId	yprovinces

Purpose	Related table	Connection field	Source table
Limits barangays to a municipality	ybarangays	munId	ymunicipalities
Associates a client with a barangay	aclients	brgyId	ybarangays
Links patient and optional parent identity to an incident	abiteinfo	patientUuid / parentUuid	aclients
Links incident details to exposure assessment	aexposure	biteInfoId	abiteinfo
Links assessment to planned and administered doses	asched	exposureId	aexposure
Provides office scope for accounts, exposures, and vaccine setup	userx; aexposure; avaccines	officeId / abtc	zofficeinfo
Supplies the selected vaccine descriptor	aexposure	vaccine selection	avaccines

Purpose	Related table	Connection field	Source table
Links report configuration and accountability records	zsignatories; bauditlogs	userxId / userId	userx
Optionally links a reported event to an existing client	ahumanrabies	patientUuid	aclients
Produces reminder information from client, exposure, and schedule data	vautoemailnotif	derived query fields	Operational records

This section retains the table-design content required by the supplied table of contents and corrects its duplicated 4.2.3 numbering to 4.2.4. The following data dictionary was generated from read-only column metadata of the connected development database. SQL type and nullability are reported as observed. PRI denotes a primary key; UNI a unique index; MUL a non-unique index. A blank key entry means no key flag was reported by the column inspection. No patient rows were required for this inventory.

aclients

Client identity and contact information, including validation state and optional PhilHealth identifier.

Table 13. aclients column definitions

Key	Null	SQL type	Field
PRI	NO	char(36)	uuid
UNI	YES	varchar(14)	philhealthNo
-	YES	varchar(30)	lname
-	YES	varchar(30)	fname
-	YES	varchar(30)	mname
-	YES	varchar(5)	suffix
-	YES	date	bdate
-	YES	varchar(6)	sex
-	YES	varchar(20)	civil
-	YES	varchar(16)	contactNo
-	YES	varchar(40)	email
-	YES	varchar(150)	employer
-	YES	int(11)	brgyId
-	YES	tinyint(1)	validated
-	YES	tinyint(1)	alive
-	YES	timestamp	created_at

abiteinfo

Incident details, patient and parent references, first-aid information, and prior immunization fields.

Table 14. abiteinfo column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id
-	YES	char(36)	parentUuid
-	YES	char(36)	patientUuid
-	YES	varchar(100)	place
-	YES	datetime	ddate
-	YES	varchar(45)	site
-	YES	varchar(10)	animal
-	YES	tinyint(1)	observable
-	YES	tinyint(1)	stray
-	YES	tinyint(1)	healthy
-	YES	tinyint(1)	immunized
-	YES	varchar(45)	owner
-	YES	varchar(45)	remarks
-	YES	tinyint(1)	labExam
-	YES	varchar(45)	result
-	YES	tinyint(1)	firstAide
-	YES	tinyint(1)	soap
-	YES	tinyint(1)	disinfectant
-	YES	tinyint(1)	tandok
-	YES	tinyint(1)	FIC

Key	Null	SQL type	Field
-	YES	tinyint(1)	tetanusToxoid
-	YES	date	givenDate
-	YES	varchar(100)	givenAt
-	YES	varchar(100)	others
-	YES	varchar(30)	prevImmunBrand
-	YES	date	prevImmunDate
-	YES	tinyint(1)	booster

aexposure

Exposure assessment, treatment-related selections, center and registration identifiers.

Table 15. aexposure column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id
UNI	NO	char(36)	uuid
-	YES	int(11)	biteInfold
-	YES	tinyint(1)	category
-	YES	varchar(100)	exposure
-	YES	tinyint(1)	activeImmun
-	YES	tinyint(1)	booster
-	YES	tinyint(1)	im

Key	Null	SQL type	Field
-	YES	int(11)	vaccine
-	YES	date	dateStarted
-	YES	int(11)	noOfDose
-	YES	tinyint(1)	passiveImmun
-	YES	int(11)	weight
-	YES	decimal(4,2)	ERIG
-	YES	decimal(4,2)	HRIG
-	YES	int(11)	regNo
-	YES	year(4)	yearCreated
-	YES	timestamp	created_at
MUL	NO	int(11)	abtc

asched

Planned and administered vaccination dates, with administration-center references.

Table 16. asched column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id
-	YES	int(11)	exposureId
-	YES	date	day0
-	YES	date	day3

Key	Null	SQL type	Field
-	YES	date	day7
-	YES	date	day14
-	YES	date	day28
-	YES	date	given0
-	YES	date	given3
-	YES	date	given7
-	YES	date	given14
-	YES	date	given28
-	YES	varchar(150)	remarks
-	YES	varchar(30)	animalstatus
MUL	YES	int(11)	given0OfficeId
MUL	YES	int(11)	given3OfficeId
MUL	YES	int(11)	given7OfficeId
MUL	YES	int(11)	given14OfficeId
MUL	YES	int(11)	given28OfficeId

avaccines

Center-associated vaccine batch descriptors and availability settings.

Table 17. avaccines column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id

Key	Null	SQL type	Field
-	YES	varchar(20)	batchno
-	YES	varchar(45)	desc
-	YES	varchar(4)	type
-	YES	date	xpiry
-	YES	int(11)	doze
-	YES	tinyint(1)	available
-	YES	tinyint(1)	active
-	NO	int(11)	officeId
-	YES	timestamp	created_at
-	YES	int(11)	created_by

zofficeinfo

Treatment-center identity, locality, operational information, and report population configuration.

Table 18. zofficeinfo column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id
-	YES	varchar(100)	alias
-	YES	varchar(10)	accro
-	YES	int(11)	mun
-	YES	varchar(70)	brgy

Key	Null	SQL type	Field
-	YES	varchar(100)	location
-	YES	int(11)	type
-	YES	int(11)	totalPopulation
-	YES	varchar(150)	operatingIn
-	YES	tinyint(1)	systemuser
-	YES	tinyint(4)	active
-	YES	tinyint(2)	sort

userx

Account identity, office, role, preferences, and authentication-related attributes.

Table 19. userx column definitions

Key	Null	SQL type	Field
PRI	NO	bigint(20) unsigned	id
UNI	NO	char(36)	uuid
-	NO	varchar(30)	lname
-	NO	varchar(30)	fname
-	YES	varchar(30)	mname
-	YES	varchar(20)	suffix
-	YES	int(11)	officeId

Key	Null	SQL type	Field
-	YES	varchar(70)	designation
UNI	NO	varchar(100)	email
-	YES	varchar(20)	contactNo
-	NO	varchar(255)	password
-	YES	varchar(255)	theme
-	YES	varchar(255)	themeColor
-	YES	varchar(100)	remember_token
-	NO	smallint(1)	active
-	YES	timestamp	created_at
-	YES	timestamp	updated_at
-	YES	varchar(2)	type
-	YES	tinyint(1)	passGenerated
-	YES	tinyint(1)	reqReset

ahumanrabies

Reported human rabies deaths, identity linkage, vaccination status, and reporting provenance.

Table 20. ahumanrabies column definitions

Key	Null	SQL type	Field
PRI	NO	bigint(20) unsigned	id
UNI	YES	char(36)	patientUuid

Key	Null	SQL type	Field
-	NO	varchar(10)	recordSource
-	NO	varchar(200)	fullName
-	YES	date	birthDate
-	YES	tinyint(3) unsigned	ageAtDeath
-	NO	varchar(10)	sex
-	YES	int(10) unsigned	provinceId
-	YES	int(10) unsigned	municipalityId
-	YES	int(10) unsigned	brgyId
-	YES	varchar(255)	provinceName
-	YES	varchar(255)	municipalityName
-	YES	varchar(255)	barangayName
-	YES	text	address
MUL	NO	date	deathDate
-	NO	varchar(40)	vaccinationStatus
-	YES	text	vaccinationNotes
-	NO	varchar(200)	reportedBy
-	NO	date	reportedDate
-	YES	varchar(200)	documentReference
-	YES	text	remarks
MUL	YES	int(11)	officeId

Key	Null	SQL type	Field
-	NO	bigint(20) unsigned	createdBy
-	NO	bigint(20) unsigned	updatedBy
-	NO	timestamp	createdAt
-	NO	timestamp	updatedAt
UNI	YES	char(64)	duplicateKey

bauditlogs

Change records identifying the target table or record, action, user, and timestamp.

Table 21. bauditlogs column definitions

Key	Null	SQL type	Field
PRI	NO	bigint(20)	id
-	YES	varchar(25)	tableName
-	YES	bigint(20)	recordId
-	YES	varchar(20)	action
-	YES	text	changes
-	YES	int(11)	userId

Key	Null	SQL type	Field
-	YES	timestamp	created_at

zsignatories

User-associated reviewer and approver information for report presentation.

Table 22. zsignatories column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id
-	NO	int(11)	userxId
-	YES	varchar(70)	reviewedby
-	YES	varchar(45)	revieweddes
-	YES	varchar(70)	approvedby
-	YES	varchar(45)	approveddes

yprovinces

Province reference values for location selection.

Table 23. yprovinces column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id
-	YES	varchar(26)	name
MUL	YES	int(11)	regionId

ymunicipalities

Municipality reference values associated with a province.

Table 24. ymunicipalities column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id
-	YES	varchar(55)	name
MUL	YES	int(11)	provinceId

ybarangays

Barangay reference values associated with a municipality.

Table 25. ybarangays column definitions

Key	Null	SQL type	Field
PRI	NO	int(11)	id
-	YES	varchar(75)	name
MUL	YES	int(11)	munId

vautoemailnotif

Database view exposing reminder payload fields; it is not a base table.

Table 26. vautoemailnotif column definitions

Key	Null	SQL type	Field
-	NO	int(11)	scheduleId
-	NO	char(36)	exposureUuid
-	YES	varchar(40)	email

Key	Null	SQL type	Field
-	YES	varchar(98)	fullname
-	NO	int(2)	scheduleDay
-	YES	date	scheduleDate
-	NO	varchar(31)	vaccinationNumber

Date fields represent either planned service dates or actual recorded administration dates; they must not be interchanged. Nullable values should preserve an unknown or not-yet-recorded state rather than silently converting it to a negative finding. The distinction matters for reporting missing information and for computing completion or overdue status.

4.3 The Extent of Compliance of the Developed System to ISO 25010 Software Quality Standards

The available evidence permits a structured discussion of quality-related implementation features, but it does not support a numerical extent-of-compliance result. No completed questionnaire dataset was supplied. The table below therefore records evidence and required evaluation for the eight ISO/IEC 25010:2011 characteristics adopted in the proposal. It is not an ISO certification statement.

Table 27. Quality evidence and remaining evaluation

Required measurement / check	Inspected evidence	Characteristic
End-to-end cases and report reconciliation	Operational pages, API routes and linked schema	Functional suitability

Required measurement / check	Inspected evidence	Characteristic
Timed tasks, concurrency and resource use	Filtered retrieval and summary endpoints	Performance efficiency
Supported-browser and integration tests	Browser frontend and API boundary	Compatibility
Observed task completion and user ratings	Structured navigation, steps and field labels	Usability
Failure recovery, duplicate and retry tests	Transactional save and schedule record locking	Reliability
Direct API access tests and privacy review	Sanctum, idle handling, role checks and audit writes	Security
Change-impact review and regression coverage	Separated pages, controllers and services	Maintainability
Clean installation and deployment reproduction	Dependency manifests and environment settings	Portability

Actual Chrome capture confirmed rendering of the sign-in, dashboard, assessment, public services, report, vaccine, and center screens included in Section 4.2.3.1. Read-only schema inspection confirmed the column definitions in Section 4.2.4. The verification did not submit a new assessment, administer a dose, delete a record, send a reminder, or execute a participant study. These limits keep the results aligned with the evidence gathered.

Table 28. Functional acceptance scenarios for pilot execution

Expected result / evidence	Scenario	ID
Correct entry or rejection; account context verified	Valid and invalid staff sign-in	F01
Clear validation; no invalid completed submission	Public pre-registration with missing / malformed data	F02
Linked client, bite, exposure and schedule saved consistently	New assessment with assigned center	F03
Duplicate rejected without partial records	Duplicate registration in the same center/year	F04
Correct exposure returned after identity check	Retrieve schedule using the intended QR identifier	F05
Defined duplicate behavior; consistent given date	Administer a scheduled dose twice	F06
Permitted undo recorded with accountability	Correct an erroneous administration	F07
Unauthorized request denied at the API boundary	Request another role's protected report	F08

Expected result / evidence	Scenario	ID
Counts match documented period and category definitions	Compare report totals with selected source records	F09
Expected timing, delivery evidence and failure handling	Process due reminders using controlled test recipients	F10
Records and essential views restored and usable	Recover a backup into an isolated environment	F11
Core tasks complete without inaccessible controls	Use keyboard and supported browsers	F12

All scenarios in the preceding table are acceptance specifications; they are not marked passed in the absence of an execution record. Formal results should include the date, tester, application version, test data reference, outcome, and defect identifier where relevant. A completed execution sheet can be added to Appendix B without changing the prescribed chapter structure.

CHAPTER V. SUMMARY OF FINDINGS AND RECOMMENDATIONS

This chapter presents the summary of findings, conclusions drawn from the results of the study, and recommendations based on the objectives of the study.

5.1 Summary of Findings

5.1.1 Objective 1: To identify the problems encountered in the existing process of animal bite case recording, immunization monitoring, and report preparation.

The study identified several concerns in the existing process of managing animal bite treatment information. These included repetitive encoding of client and treatment information, difficulty in retrieving previous vaccination records, challenges in monitoring scheduled vaccination doses, and the additional effort required to consolidate animal bite and rabies-related information for provincial reporting.

The existing process also presented difficulties in maintaining continuity of information when a patient receives succeeding vaccine doses at another participating treatment center. These findings served as the basis for identifying the functional and information requirements of the developed system.

5.1.2 Objective 2: To design and develop an Integrated Management System.

The Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center was successfully developed as a web-based application.

The developed system provides functions for online client pre-registration, client and exposure record management, animal bite case assessment, vaccination scheduling and monitoring, recording of administered doses, cross-center continuation of treatment, QR-based record retrieval, vaccination-history verification, treatment-center and vaccine availability information, human rabies recording, automated email reminders, vaccination-slip generation, dashboard monitoring, and report generation.

The system also provides administrative functions for user account management, treatment-center maintenance, signatories, application preferences, provincial monitoring, and Department of Health reporting.

5.1.3 Objective 3: To develop a centralized data management structure for animal bite and rabies-related information.

The developed system uses a centralized database to maintain related client, guardian, animal bite, exposure, vaccination, treatment-center, user, human rabies, and reporting information.

The centralized structure allows authorized users to retrieve relevant client and vaccination records across participating treatment centers while maintaining restrictions on the modification of records according to user role and treatment-center ownership.

The system also maintains relationships between exposure records, vaccination schedules, administered doses, and the centers responsible for providing the service. This structure supports continuity of vaccination and the consolidation of information required for center-level and provincial reporting.

5.1.4 Objective 4: To evaluate the software quality of the developed system using ISO/IEC 25010:2011.

The developed system was evaluated using the eight software quality characteristics of ISO/IEC 25010:2011, namely functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

A total of 20 respondents from the Cagayan Provincial Health Office participated in the evaluation, consisting of 10 IT experts, one doctor, three nurses, one midwife, one nursing attendant, and four administrative staff.

This subsection should contain your actual computed ISO 25010 results once available. For example, this is where you will state the overall mean and the

descriptive interpretation of each characteristic. Do not put "Very Great Extent" or similar conclusions until you have the actual results.

5.1.5 Objective 5: To formulate recommendations based on the findings of the study.

Based on the evaluation, testing, and observations of the respondents, recommendations were identified for the continued improvement and implementation of the Integrated Management System. These recommendations focused on system refinement, user orientation, report validation, access control, reminder delivery, system maintenance, and future enhancements.

5.2 Conclusions

Based on the findings of the study, an Integrated Management System was developed to address the identified concerns in animal bite case recording, vaccination monitoring, record retrieval, treatment continuity, and provincial reporting.

The system provides a centralized environment for managing client and exposure information, vaccination schedules, administered doses, treatment-center records, human rabies information, and reporting data. It also supports cross-center continuation of vaccination, public access to selected vaccination information, QR-based record retrieval, vaccination slips, and automated email reminders.

The development and initial implementation of the system demonstrate that the major functional requirements identified during the study were incorporated into the application. The Provincial Health Office has begun testing the system and providing recommendations for further refinement prior to wider implementation among participating treatment centers.

5.3 Recommendations

1. The Cagayan Provincial Health Office should continue the testing and refinement of the Integrated Management System and incorporate verified recommendations from the participating PHO personnel.
2. Orientation and training should be provided to personnel from participating PLGU, MLGU, RHU, and Animal Bite Treatment Centers before full implementation of the system.
3. Generated center-level, provincial, and Department of Health reports should be regularly validated to ensure the accuracy of classifications, computations, reporting periods, and consolidated information.
4. The automated vaccination reminder service should be continuously monitored to ensure that scheduled notifications are properly processed and delivered to patients with valid email addresses.
5. User roles, access permissions, and treatment-center restrictions should be periodically reviewed to maintain appropriate protection of client and treatment information.
6. Regular database backup, restoration testing, server monitoring, and system maintenance should be conducted to support continuous operation and data protection.
7. The system should be continuously enhanced based on operational requirements, user feedback, and changes in reporting or animal bite treatment procedures.
8. Future enhancements may include more detailed vaccine inventory management, additional notification methods, stronger public identity verification, and expanded monitoring and reporting functions.

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APPENDICES

Appendix A. Request Letter for Gathering Data

February 3, 2026

GEN. EDGAR B. AGLIPAY (RET.)
Governor
Provincial Government of Cagayan
Alimanao Hills, Tuguegarao City, Cagayan

Dear Governor:

The undersigned is a student of the Graduate School taking up Master of Science in Information Technology at Cagayan State University, Carig Campus.

As a requirement of the course the undersigned respectfully requests permission from your good office to conduct a study entitled, **"Integrated Animal Bite Treatment Center Information System with QR-Based Vaccination Schedule Retrieval and Automated E-mail Notifications"** at your institution.

Please be assured that the data I gather will be used solely for the purpose of my thesis. I would be very grateful for your favorable consideration of this request.

Thank you very much for your time and kind attention.

Respectfully yours,

MAVERICK M. BALLESTEROS
Student, MSIT

Appendix B. Request for Pilot Testing

REQUEST FOR PILOT TESTING AND SYSTEM EVALUATION

September 21, 2026

SIR SHAMON C. DE YRO
Nurse III
Animal Bite Treatment Center
Cagayan Provincial Health Office
Province of Cagayan

Dear Sir:

The undersigned is a student of the Graduate School taking up Master of Science in Information Technology at Cagayan State University, Carig Campus.

In connection with the conduct of the study entitled **"Development of an Integrated Management System for the Cagayan Provincial Health Office – Animal Bite Treatment Center,"** the undersigned respectfully requests permission to conduct pilot testing and system evaluation at the Animal Bite Treatment Center of the Cagayan Provincial Health Office.

The activity will include system orientation, demonstration and testing of the major functions of the developed system, observation of system performance, and administration of the software quality evaluation questionnaire based on ISO/IEC 25010:2011.

The pilot testing aims to determine whether the developed system supports the intended processes for client information management, animal bite assessment, vaccination scheduling and monitoring, record retrieval, treatment continuity, reporting, administrative functions, and other related operations of the Animal Bite Treatment Center.

Comments, observations, and recommendations from the participants will also be gathered and used as basis for the further improvement and refinement of the developed system.

Please be assured that all information gathered during the activity will be treated with confidentiality and will be used solely for academic and research purposes. Demonstration or test data may be used whenever necessary to avoid unnecessary disclosure of actual patient information.

Your favorable consideration and approval of this request will be highly appreciated.

Thank you very much for your time and kind attention.

Respectfully yours,

MAVERICK M. BALLESTEROS

Researcher / MSIT Student

Appendix C. Users Manual

DOCUMENT CONTROL

Table 31. Users manual document control

Controlled information	Item
IMS Users Manual	Document title
Integrated Management System for the Cagayan Provincial Health Office - Animal Bite Treatment Center	System
Presentation draft - September 2026	Version
Maverick M. Ballesteros	Prepared by
Cagayan State University - Graduate School, Carig Campus	Institution
Authorized PHO, ABTC, RHU, and system-administration personnel	Intended users
For adviser, panel, and authorized office review	Review status

This manual follows the implemented browser routes and the current application workflow. Labels can vary with the account role and subsequent application changes. An authorized account and the institution's assigned system address are required for staff functions. The localhost address in screenshots identifies the documentation environment only.

C.1 Access and sign-in

Staff access establishes the account and office context used in operational requests.

1. Open the institution's assigned application address and select the sign-in page.
2. Enter the assigned email address and password. Confirm that the displayed user and office are correct.
3. Complete a required personal-password change when prompted. Do not continue with an account assigned to the wrong center.
4. When a session expires, sign in again and verify whether any unsaved work must be re-entered.
5. Use the account logout action when finished, especially on a shared workstation.

If sign-in fails, confirm the entered identifier and use the institution's reset process. Do not ask another staff member to share credentials. An account with incorrect office information should be corrected by an authorized administrator.

C.2 Public pre-registration

Pre-registration collects identity information before professional assessment.

1. Open Online Services and choose Pre-Registration.
2. Enter the required demographic, contact, and locality information. Review spelling and dates.
3. Use the dependent or parent fields only when applicable to the person being registered.
4. Review the information and submit once. Keep any confirmation or retrieval information displayed.
5. At the center, ask staff to verify the submitted record before creating a new identity record.

Submission does not complete an exposure assessment, reserve treatment, or certify eligibility for a benefit. If an error occurs, verify whether submission succeeded before repeating it. Assisted registration should remain available.

C.3 Public history verification and center information

Public services offer a limited retrieval and information path.

A displayed record may be incomplete if a service was not encoded. An availability indicator is not a guaranteed stock reservation. Errors in public content should be reported to the responsible office.

C.4 Dashboard

The dashboard supports operational review for a chosen reporting scope.

1. Open Dashboard from the navigation menu.
2. Select the year, quarter or whole year, and center where the account permits it.
3. Read the labels for each measure before comparing values. Pending doses and cases are different units.
4. Review the category, demographic, and source summaries.
5. Use the corresponding operational view to investigate a discrepancy through an authorized workflow.

Do not interpret demonstration values as official provincial statistics. If a value is unexpected, confirm filters and the source-record date basis before changing records.

C.5 Case assessment

Case assessment records a new incident and exposure using a verified client identity.

1. Open Case Assessment and check the office shown in the workspace.
2. Identify the client, using an existing record where appropriate. Confirm name, birth date, contact details, and dependent information.
3. Complete the incident fields, including date, location, site, and animal-related details where available.
4. Record the exposure category and treatment-related selections based on the authorized professional assessment.
5. Review vaccination-related fields and the planned dates. Do not mark a dose as given merely because it is scheduled.

6. Review the complete entry and save once. Confirm the returned registration and retrieval information.

If validation fails, correct the named field and review related values. The application requires an assigned office to save an assessment. A duplicate registration should be resolved through the authorized correction process, not by creating an arbitrary conflicting number.

C.6 Rabies immunization history

The immunization view supports review of recorded exposure and vaccination information.

1. Open Rabies Immunization.
2. Search or filter for the intended client or registration.
3. Verify the person and exposure before opening or editing the record.
4. Review active or previous vaccination details and compare them with the available clinical documentation.
5. Use permitted update actions only to record verified information, then confirm the saved values.

A single client may have more than one exposure episode. Avoid modifying an older episode when documenting a new incident. Conflicting histories should be escalated to the responsible clinician or data custodian.

C.7 QR retrieval and dose administration

The schedule workflow links a retrieval identifier to planned and completed doses.

1. Open Schedule and use the available registration search or QR retrieval control.
2. Allow camera access only when using the scanning function on the approved device.
3. Verify the retrieved identity and episode; a QR match alone does not establish identity.
4. Select the intended scheduled dose and record the date actually administered.

5. Confirm the service center and save the administration record. Review the updated given date.

6. Use Undo only to correct a documented mistake and follow the office's accountability procedure.

If scanning fails, use the available identifier lookup and inspect the printed code. If the dose is already recorded, verify the existing entry before attempting another update. Escalate conflicting dates.

C.8 Client records

Client records provide reusable identity information for incident and exposure documentation.

1. Open Clients and locate the person using the available search controls.
2. Compare more than one identifying field before choosing the record.
3. Review demographic information and distinguish the patient from an associated parent or dependent.
4. Refer duplicates or identity corrections to the authorized staff workflow.
5. Return to assessment or history with the correct client context.

Do not infer a missing history from a failed name search alone. Spelling, suffixes, dates, and identifiers can affect matching. Corrections should preserve the relationship to existing exposure records.

C.9 Human rabies records

The human rabies workflow documents reported events and their provenance.

1. Open Human Rabies using an authorized account.
2. Choose an existing patient where applicable or follow the supported external-record option.
3. Record the event date, demographic details, reported vaccination status, and available notes.
4. Enter who reported the event, the reporting date, and the supporting document reference.
5. Review the record carefully and save through the permitted action.

A record in this module reflects information entered into the system; it is not itself laboratory confirmation. Keep reported status and uncertainty explicit. Deletion or correction should follow the office's record-governance process.

C.10 Accomplishment reports

The report summarizes the selected center and date interval.

1. Open Accomplishment Report.
2. Choose the From and To dates and confirm that the range is correct.
3. Review the displayed center and summary headings.
4. Use Preview PDF to inspect the report layout before downloading.
5. Check signatory details, dates, totals, and report scope before official submission.

When numbers differ from a manual tally, compare the same record population and date definition. Do not change source records solely to force agreement with an unrelated total.

C.11 Bite cases and cohort reports

Administrator reporting supports municipality summaries and template output.

1. For Bite Cases, select the year and quarter or whole-year period.
2. Review municipality rows and totals before generating the PDF.
3. For Cohort Report, select the required year and choose Download Excel.
4. Open the workbook and verify the center list, population fields, quarter sections, and any fields requiring further completion.
5. Retain the approved output according to the office's reporting procedure.

The cohort interface is described as a template-preparation function. Confirm which indicators are computed before treating the workbook as a fully completed submission.

C.12 Vaccine setup

Vaccine setup maintains the records available to the clinical workflow.

1. Open Vaccines and choose the applicable availability filter.
2. Review batch number, description, type, expiry, and the current in-use marker.
3. Use Add New or permitted update actions to record an authorized vaccine entry.
4. Change availability or in-use status only after confirming the real operational state.
5. Recheck the displayed selection before using it in an assessment.

Availability flags do not replace a stock ledger, expiry checks, or physical inventory. The responsible personnel must verify that the selected product and batch are appropriate.

C.13 Centers, accounts, and signatories

Administrative setup supplies reference information used throughout the application.

1. In Centers, locate the facility and verify its name, alias, locality, operating information, and population configuration.
2. In Accounts, assign the correct office, account type, and active state using an authorized administrator account.
3. Handle reset and generated-password actions through the approved account-management procedure.
4. In Signatories, enter the reviewer and approver names and designations used for the account's report outputs.
5. Preview an affected report or workflow after a configuration change to confirm its presentation.

Administrative privileges should match assigned responsibilities. Account impersonation or access to generated credentials is privileged functionality and should be governed and reviewed by the institution.

C.14 Daily use and incident reporting

Consistent operation helps maintain trustworthy records.

1. At the start of work, confirm the account, office, and application availability.
2. Resolve incomplete records through the appropriate responsible person rather than entering assumed facts.
3. When an error appears, record the time, screen, action, and non-sensitive error message.
4. Report the issue through the designated support contact. Include an identifier only through an approved secure channel.
5. At the end of work, complete authorized updates and sign out.

Do not repeatedly submit an uncertain transaction. Check whether it completed before retrying. An operational outage requires the office's continuity process; the application has no verified automatic offline synchronization.

Deployment and maintenance handover

The technical custodian should retain the frontend and backend source, dependency lock files, complete database schema and view definitions, approved environment configuration, and report templates. A clean environment must be able to recreate the application without depending on undocumented tables that exist only in the development database. Secrets and personal data do not belong in the public source-code appendix.

The deployment procedure should configure a supported runtime, database connection, application key, frontend API address, authentication settings, report binary, scheduler, and optional realtime service. Use a dedicated test recipient to verify the reminder chain under explicit test authorization. Record the deployed version and perform a backup-restoration exercise before treating backups as dependable.

Table 32. Troubleshooting guide

Appropriate next step	Check	Symptom
Use reset/support process; do not share accounts	Credentials, account active state, API availability	Cannot sign in
Sign in and confirm unsaved work	Idle-session policy and login state	Session expired
Verify identity with authorized staff	Identifiers, spelling, date and filters	No search match
Reconcile filters before assuming missing cases	Date range, center scope, source-record dates	Report empty
Technical custodian checks logs and binaries	Report service and wkhtmltopdf configuration	PDF generation fails
Inspect controlled delivery records	Address, due-view selection, scheduler and endpoint	Reminder missing
Use identifier lookup where available	Camera permission, lighting and supported browser	QR scanner unavailable

Appendix D. ISO 25010 Questionnaire

Survey-Questionnaire

Software Effectiveness Questionnaire of the System using ISO/IEC 25010:2011 or ISO 9126 Quality Standards

(This survey-questionnaire is designed to determine the efficiency of the _____).

Part I. Profile of the Respondents

Kindly fill-in the blanks with required information or by simply putting a (√) check mark.

Age: __ below 30 years old
__ 31 – 40 years old
__ 41 – 50 years old
__ 51 – 60 years old
__ above 60 years old

Sex: ___ Male ___ Female

Highest Educational Attainment:
College ___
Post-Graduate ___

Part II. Assessment of the Extent of Compliance to ISO 25010:2011 Software Quality

Instruction: Kindly indicate by putting a check mark (√) in the appropriate box your assessment on the extent of compliance to ISO 25010:2011 of the developed *web-based performance measurement system*in terms of Functionality, Reliability, Usability, Efficiency, Maintainability, Portability and security using the following scale:

5 – Compliant to a very high extent

4 – Compliant to a high extent

3 – Compliant to a moderate extent

2 – Compliant to a low extent

1 – Compliant to a very low extent

A. Functionality	**5**	**4**	**3**	**2**	**1**	**Comments**
1. The required task is performed by the system satisfactorily						
2. The output generated by the system is correct.						
3. The system helps the user overcome any problem encountered while using it.						
4. The system regulates user access.						
B. Reliability	**5**	**4**	**3**	**2**	**1**	**Comments**
5. Most errors have been corrected over time.						
6. The system is capable of maintaining its performance despite the presence of errors during operations.						
7. The system provides restore operation of lost data after a failure.						
C. Usability	**5**	**4**	**3**	**2**	**1**	**Comments**
8. The user interface is intuitive and receptive to learning.						
9. The information generated by the system is easy to understand and act upon.						
10. The documentation is very informative.						
11. The system performs tasks in a straightforward manner.						
D. Efficiency	**5**	**4**	**3**	**2**	**1**	**Comments**
12. The system responds immediately to user commands other input.						
13. The system maximizes the utilization of available resources (i.e. people, data, machines, and materials) efficiently.						
14. The system performs efficiently on						

intended tasks with ease and scalability.						
Maintainability	**5**	**4**	**3**	**2**	**1**	**Comments**
Deficiencies can be easily diagnosed and modified from the system.						
The system can be easily customized and modified.						
The system continues functioning if changes are made.						
The system can easily be tested.						
F. Portability	**5**	**4**	**3**	**2**	**1**	**Comments**
The system performs its functions efficiently even it is moved to another environment.						
The system can be installed easily.						
The system can be installed easily.						
The system complies with the portability standards.						
The system can take the place of other software.						
Compatibility	**5**	**4**	**3**	**2**	**1**	**Comments**
The system can perform its required functions efficiently						
The degree to which the system can exchange information and use the information that has been exchange,						
Security	**5**	**4**	**3**	**2**	**1**	**Comments**
The system has its security level access with pre-determined action.						
The data is secured as well as its associated records.						
Access to top-level management data is topped with necessary security credentials.						

Source: ISO 25010:2011 Software Quality Model

Kindly list below YOUR PERCEIVED "BEST FEATURES OR STRENGTH" of the developed system.

1. _____
2. _____
3. _____
4. _____
5. _____

Kindly list below YOUR PERCEIVED "LIMITATIONS OF THE SYSTEM NEEDED TO BE ADDRESSED".

6. _____
7. _____
8. _____

9. _____
10. _____

Suggestions:

_____

_____

_____

Appendix E. Source Code

TECHNICAL SOURCE APPENDIX

This appendix identifies the principal implementation files and includes selected verbatim excerpts from the inspected project. The complete repositories remain the implementation deliverable. Environment secrets, credentials, patient records, generated tokens, and dependency source are excluded from the manuscript.

Table 43. Source-code organization

Responsibility	Location
Page routes, session validation and administrator route metadata	src/router/index.ts
Structured assessment interface	src/pages/Assesment.vue
Public online services and pre-registration	src/pages/PreRegistration.vue
Case history and vaccination follow-up	src/pages/Immunization.vue; Schedule.vue
Reported human rabies records	src/pages/HumanRabies.vue
Report filters, previews and download controls	src/pages/*Report.vue
API route and middleware definitions	routes/api.php
Client, assessment, dashboard and schedule operations	app/Http/Controllers/ AssessmentController.php
Report queries and output generation	app/Http/Controllers/ ReportController.php

Responsibility	Location
Reminder selection and external request	app/Services/ VaccineReminderService.php
Scheduled reminder command	app/Console/Kernel.php
Versioned schema changes included in the repository	database/migrations

E.1 Reminder selection and delivery

Source: app/Services/VaccineReminderService.php, lines 1-65.

```
<?php
namespace App\Services;

use Illuminate\Support\Facades\DB;
use Illuminate\Support\Facades\Http;

class VaccineReminderService
{
    public function dueReminders()
    {
        return DB::table('vautoemailnotif')
            ->select([
                'scheduleId',
                'exposureUuid',
                'email',
                'fullname',
                'scheduleDay',
                'scheduleDate',
                'vaccinationNumber',
            ])
            ->whereNotNull('email')
```

```
            ->where('email', '<>', '')
            ->get();
    }

    public function send(object $reminder): bool
    {
        try {
            $response = Http::acceptJson()->connectTimeout(10)->timeout(30)
                ->withOptions(['allow_redirects' => false])
                ->post(config('services.abtc_reminders.endpoint'), [
                    'email' => $reminder->email,
                    'fullname' => $reminder->fullname,
                        'scheduleDate' => $this->formatScheduleDate($reminder-
>scheduleDate),
                    'vaccinationNumber' => $reminder->vaccinationNumber,
                    'exposureUuid' => $reminder->exposureUuid,
                ]);

            return $response->successful();
        } catch (\Throwable $exception) {
            return false;
        }
    }

    public function sendDueReminders(): array
    {
        $sent = 0;
        $failed = 0;

        foreach ($this->dueReminders() as $reminder) {
            if ($this->send($reminder)) {
```

```
                $sent++;
            } else {
                $failed++;
            }
        }

        return ['sent' => $sent, 'failed' => $failed];
    }
    private function formatScheduleDate($value): ?string
    {
        if ($value instanceof \DateTimeInterface) {
            return $value->format('Y-m-d');
```

E.2 Scheduled execution

Source: app/Console/Kernel.php, lines 10-19.

```
    protected function schedule(Schedule $schedule): void
    {
        $schedule->command('abtc:send-vaccine-reminders')
            ->dailyAt(config('services.abtc_reminders.time', '18:00'))
            ->withoutOverlapping();
    }
```

E.3 Administrator report authorization

Source: app/Http/Controllers/ReportController.php, lines 12-18.

```
class ReportController extends Controller
{
    private function authorizeAdministrator(Request $request): void
    {
        abort_unless(in_array($request->user()?->type, ['SA', 'A'], true), 403, 'Only administrators can access this report.');
    }
```

CURRICULUM VITAE

Personal Data



Full Name:	**MAVERICK M. BALLESTEROS**
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Professional Profile

An information technology professional with experience in systems analysis, software development, web application development, desktop application development, database management, systems maintenance, and government information systems. Currently serving as Computer Maintenance Technologist I and designated as Lead Programmer of the Information Systems Division under the Office of the Provincial Administrator, Provincial Government of Cagayan.

Experienced in analyzing user and organizational requirements, designing and developing information systems, maintaining databases, and implementing both web-based and desktop applications to support government operations. Also involved in the maintenance, enhancement, deployment, and technical support of existing information systems.

Socials

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